Prismatic Method Analysis (Mark 2): Global AMR Response 2015–2026

A Methodological Validation Study for R. Jazinski's Silent Transitions Framework

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Section 0: Pre-Analysis Reflexive Declaration

This analysis applies the Prismatic Method (Mark 2) to the global antimicrobial resistance (AMR) response spanning 2015–2026, a period bookended by the WHO Global Action Plan (2015) and the present moment of accumulated implementation experience. The analyst enters this field with specific epistemic commitments and limitations that must be declared before signal processing begins.

Epistemic position: I approach this corpus as a methodological validation exercise for R. Jazinski's Silent Transitions framework, which posits that major sociotechnical transformations often proceed through mechanisms invisible to dominant governance logics. My training emphasizes pattern recognition across heterogeneous evidence types, but I carry inherent biases toward coherence-seeking and may underweight genuinely contradictory or incommensurable signals. I am predisposed to find structure where fragmentation may be the more accurate characterization.

Corpus limitations: The signal set derives from 309 deduplicated papers retrieved through systematic database searches (SciSpace, Google Scholar, PubMed) with explicit focus on policy, governance, epidemiology, and One Health dimensions of AMR response. This corpus over-represents published academic and policy literature in English, creating systematic blind spots: (1) implementation realities documented in grey literature, national reports, and non-English sources; (2) industry perspectives on R&D decisions and market exits; (3) frontline clinical and veterinary practice patterns; (4) community-level resistance dynamics in under-surveilled regions. The signal set is strongest on global governance architecture and weakest on ground-truth implementation and microbial evolutionary dynamics.

Temporal framing: The 2015–2026 window captures a distinct policy era: post-Global Action Plan but pre-any binding international treaty. This eleven-year span allows assessment of first-generation National Action Plan (NAP) implementation but may be too short to detect slow-moving epidemiological or evolutionary signals. The analysis necessarily treats 2026 as "present," aware that surveillance lags mean 2024 data may be the most recent available for many indicators.

Methodological commitments: The Prismatic Method requires processing identical signals through multiple interpretive grammars to expose how framing shapes closure. I commit to: (1) treating signals as theory-laden observations rather than raw facts; (2) maintaining fidelity to each grammar's internal logic even when it produces uncomfortable conclusions; (3) resisting premature synthesis that would collapse productive tensions between grammars; (4) foregrounding counter-signals and leakage as epistemically valuable rather than noise to be explained away.

Anticipated tensions: I expect this analysis will surface conflicts between: technocratic governance optimism (NAPs as progress) versus implementation realism (isomorphic mimicry); global burden quantification (5 million deaths) versus attribution uncertainty; One Health rhetoric versus sectoral fragmentation; market failure diagnosis versus political economy of pharmaceutical capital. The method's value lies in holding these tensions open rather than resolving them prematurely.

Reflexive limitation: As an AI system, I lack embodied experience of AMR's clinical, agricultural, or lived realities. My analysis operates entirely within textual representations produced by specific epistemic communities (academic researchers, WHO officials, policy analysts). This creates a systematic bias toward governance-legible phenomena and away from ungoverned or informally governed resistance dynamics.

This declaration establishes the interpretive ground from which signal processing proceeds. The analysis that follows is not a view from nowhere but a situated reading of a particular corpus through a specific methodological lens, undertaken for explicit validation purposes within Jazinski's theoretical framework.

Section 1: Observation Field Declaration

Field designation: Global antimicrobial resistance governance and response architecture, 2015–2026

Temporal boundaries: May 2015 (WHO Global Action Plan adoption at WHA68) through March 2026 (present analysis date), encompassing eleven years of first-generation NAP implementation, surveillance network expansion, and evolving governance experimentation.

Spatial scope: Global in aspiration, with empirical concentration in: (1) WHO member states with documented NAP development (>100 countries); (2) GLASS surveillance network participants (variable coverage); (3) high-income countries with established stewardship programs; (4) LMIC regions with documented implementation gaps (sub-Saharan Africa, South/Southeast Asia). Systematic under-representation of conflict-affected states, small island nations, and regions with minimal laboratory capacity.

Phenomenon under observation: The sociotechnical assemblage constituted by: international policy coordination mechanisms (WHO/FAO/WOAH tripartite, later Quadripartite with UNEP); national governance responses (NAPs, stewardship programs, surveillance systems); pharmaceutical R&D and market dynamics; agricultural antibiotic use patterns; clinical resistance trends; and the complex feedback loops among these domains. The field encompasses both formal governance architecture and its implementation realities, including documented gaps, failures, and unintended consequences.

Signal sources: 309 deduplicated academic papers and policy documents retrieved through systematic search across multiple databases, supplemented by extracted empirical signals including: mortality/morbidity estimates (5 million AMR-associated deaths, 2019); regional resistance disparities (3-4× higher rates in LMICs); NAP adoption patterns (>100 countries); stewardship effectiveness data (32% resistance reduction in implementing facilities); agricultural use patterns; R&D pipeline constraints; economic burden models ($100-150B annual); surveillance coverage gaps; and documented counter-signals (gonorrhea resistance, isomorphic mimicry, industry exits).

Observational instruments: The corpus itself functions as the primary instrument, shaped by: database search algorithms and ranking logics; academic publication and peer review filters; English-language bias; policy document accessibility; and the specific search terms deployed (AMR, global response, governance, One Health, stewardship, surveillance). These instruments systematically privilege certain signal types (published quantitative estimates, policy declarations) while attenuating others (implementation failures, informal practices, evolutionary dynamics in under-surveilled regions).

Field boundaries and exclusions: This observation field explicitly excludes: (1) detailed microbial genomics and evolutionary mechanisms (treated as black-boxed drivers); (2) pharmaceutical industry internal decision-making (accessible only through external policy analysis); (3) farm-level and clinical practice patterns (except as aggregated in surveillance or survey data); (4) patient and community perspectives on antibiotic access and use; (5) environmental AMR reservoirs and transmission pathways (acknowledged in

One Health rhetoric but empirically thin in corpus). The field is bounded by governance-legible phenomena and systematically misses ungoverned or informally governed dynamics.

Epistemic status of observations: Signals in this field are not raw data but theory-laden constructs produced by specific measurement regimes: WHO surveillance networks define what counts as resistance; burden-of-disease models embed attribution assumptions; NAP assessments operationalize "implementation" through particular indicators; economic models project costs using contested parameters. Each signal carries the epistemic commitments of its producing institution. The observation field is thus a constructed space shaped by the governance architectures it purports to describe—a reflexive entanglement that the Prismatic Method must navigate carefully.

Field dynamics: The 2015–2026 period exhibits several characteristic dynamics: (1) rapid policy diffusion (NAP adoption) followed by slower implementation; (2) expanding surveillance coverage with persistent geographic gaps; (3) growing burden quantification alongside attribution debates; (4) sustained R&D market failure despite policy attention; (5) One Health rhetorical consensus with operational fragmentation. These dynamics suggest a field in which symbolic governance moves faster than material transformation—a pattern central to Silent Transitions analysis.

This observation field is not a neutral container of facts but an actively constructed domain shaped by the governance logics, measurement regimes, and epistemic communities that produce its signals. The Prismatic Method proceeds by processing these constructed signals through multiple interpretive grammars to expose how different framings generate different closures around the "same" empirical terrain.

Section 2: Signal Set

2.1 Core Signals (10-15 primary empirical patterns)

Signal 1: Global mortality burden quantification Approximately 5 million deaths associated with bacterial AMR in 2019, established through large-scale burden-of-disease modeling synthesizing surveillance data, hospital records, and statistical attribution methods [1], [2]. This signal represents the most widely cited quantitative anchor for AMR's global health impact, functioning as a governance-legible metric that translates microbial resistance into mortality statistics comparable to other disease burdens.

Signal 2: Regional resistance disparity Resistance rates and associated mortality burden 3-4× higher in low- and middle-income countries compared to high-income settings, documented across multiple pathogen-drug combinations in GLASS and regional surveillance networks [2]. This signal reveals AMR as a phenomenon of global health inequality, with excess burden concentrated in regions with weakest governance capacity, laboratory infrastructure, and stewardship implementation.

Signal 3: National Action Plan diffusion Over 100 WHO member states completed or initiated NAP development following the 2015 Global Action Plan, representing rapid policy diffusion across diverse governance contexts [1], [5], [6]. This signal demonstrates successful norm transmission through international health governance architecture, with NAPs serving as the primary national-level policy instrument for AMR response.

Signal 4: Vertical-horizontal alignment gap Strong vertical alignment of NAPs with WHO Global Action Plan objectives but substantially weaker horizontal coordination across sectors and lower implementation fidelity, particularly in resource-constrained settings—a pattern characterized as "isomorphic mimicry" where policy documents exist without operational transformation [4], [5], [7]. This signal exposes the gap between policy adoption and implementation, suggesting symbolic compliance may substitute for substantive action.

Signal 5: Stewardship effectiveness in implementing sites Antimicrobial stewardship programs reduced resistance rates by approximately 32% in facilities with documented implementation, demonstrating technical efficacy of stewardship interventions where operationalized [2]. This signal provides evidence for stewardship as an effective intervention while simultaneously highlighting that effectiveness is conditional on implementation—a distinction often elided in policy discourse.

Signal 6: Surveillance coverage gaps WHO GLASS and regional networks exhibit uneven reporting, with many LMICs showing limited laboratory capacity and systematic under-reporting, producing major geographic blind spots in global resistance monitoring [10], [11]. This signal reveals that global surveillance architecture remains incomplete, with data quality and coverage inversely correlated with resistance burden—a perverse pattern where least is known about most-affected regions.

Signal 7: Sub-Saharan Africa preparedness deficit AMR preparedness scores in sub-Saharan Africa substantially below overall JEE (Joint External Evaluation) means, reflecting laboratory capacity, infection prevention and control, and NAP implementation shortfalls [14]. This signal identifies SSA as a region where multiple governance deficits compound, creating systematic vulnerability to AMR impacts and limited capacity for response implementation.

Signal 8: Agricultural antibiotic use parity or excess Antibiotic consumption in agricultural sectors equals or exceeds human medical use in many countries, with agricultural use identified as major AMR driver but subject to weaker regulatory oversight, particularly in LMICs [3], [29], [30]. This signal highlights the One Health dimension of AMR while exposing governance asymmetry between human and animal health sectors.

Signal 9: Pharmaceutical R&D decline and market failure Sustained decline in conventional pharmaceutical investment in antibacterial R&D despite policy attention, with documented industry exits and consolidation, driven by fundamental market failures (low returns, stewardship-constrained sales) that existing incentive structures fail to correct [9], [21]. This signal reveals a structural economic contradiction: stewardship aims to preserve antibiotic effectiveness by limiting use, but limited use undermines commercial viability, creating a market failure that voluntary mechanisms cannot resolve.

Signal 10: Economic burden modeling Recent syntheses estimate $100-150 billion annual economic costs attributable to AMR, with projections of substantially larger long-term macroeconomic losses without effective intervention, used to justify policy investment and international coordination [2], [9]. This signal translates microbial resistance into economic terms legible to finance ministries and development institutions, though burden estimates embed contested attribution assumptions and modeling choices.

Signal 11: One Health policy-practice gap One Health principles embedded in global guidance and most NAPs, with strong rhetorical consensus, but operational integration across human, animal, and environmental health sectors remains incomplete, particularly in multisectoral surveillance and coordinated stewardship [4], [7], [9], [24]. This signal exposes tension between One Health as governance ideal and the sectoral fragmentation of actual implementation.

Signal 12: Priority pathogen list updates WHO Bacterial Priority Pathogens List updated in 2024, identifying critical, high, and medium priority pathogens for R&D focus, with modest flow of new agents targeting priority pathogens approved post-2017 [20]. This signal demonstrates continued governance attention to pipeline needs while implicitly acknowledging that R&D output remains insufficient relative to identified priorities.

Signal 13: Gonorrhea resistance sentinel Documented emergence of ceftriaxone- and azithromycin-resistant Neisseria gonorrhoeae in global surveillance, representing rapid clinically consequential resistance despite stewardship efforts and serving as sentinel for treatment failure risk [19]. This signal functions as counter-evidence to stewardship optimism, demonstrating that resistance evolution can outpace governance response even for closely monitored pathogens.

Signal 14: Resource constraint and competing priorities Systematic documentation that resource scarcity, weak health systems, and competing health priorities constrain operational uptake of stewardship, surveillance, and animal-sector reforms in LMICs, with Covid-19 pandemic further straining implementation capacity [11], [14], [24], [30]. This signal contextualizes implementation gaps within broader political economy of global health, where AMR competes with other urgent priorities in resource-limited settings.

Signal 15: Governance instrument debates Ongoing contestation over whether binding international legal instruments (e.g., Framework Convention) are necessary versus strengthened non-binding coordination, reflecting deeper disagreements about sovereignty, enforcement, and governance design [11], [12]. This signal reveals that global governance architecture remains unsettled, with fundamental questions about coordination mechanisms still contested among key actors.

2.2 Counter-Signals (3-5 patterns contradicting dominant narratives)

Counter-signal 1: Isomorphic mimicry and policy-practice disconnect Documented cases where NAPs exist with strong textual alignment to global frameworks but operational implementation lags substantially, suggesting symbolic policy adoption may substitute for substantive governance transformation—contradicting narratives of NAP adoption as evidence of effective response [4], [5], [7]. This counter-signal challenges the equation of policy diffusion with implementation progress.

Counter-signal 2: Persistent industry exit despite incentive attention Continued pharmaceutical company withdrawal from antibacterial R&D despite sustained policy focus on push/pull incentives, subscription models, and de-linkage proposals, suggesting existing incentive designs fail to overcome fundamental market structure problems [9], [21]. This counter-signal contradicts optimistic narratives that incremental incentive reforms will revive the antibiotic pipeline.

Counter-signal 3: Surveillance expansion with persistent blind spots Growth in GLASS participation and surveillance network coverage alongside persistent and systematic under-reporting from highest-burden regions, suggesting surveillance expansion may increase data volume without proportionally reducing geographic gaps or improving data quality where most needed [10], [11], [14]. This counter-signal challenges narratives of surveillance progress by highlighting that expansion does not equal equity or adequacy.

Counter-signal 4: Stewardship effectiveness versus coverage Strong evidence for stewardship effectiveness in implementing facilities alongside documentation that stewardship coverage remains limited, particularly in LMICs and agricultural sectors, suggesting demonstrated efficacy may not translate to population-level impact if implementation remains sparse [2], [11], [24]. This counter-signal exposes the gap between intervention efficacy and public health effectiveness at scale.

Counter-signal 5: One Health consensus without integration Near-universal rhetorical endorsement of One Health principles in policy documents alongside persistent operational fragmentation across human, animal, and environmental health sectors, suggesting consensus on principles does not generate institutional integration [4], [7], [9], [24], [29], [30]. This counter-signal reveals that ideational convergence may mask continued sectoral silos in practice.

2.3 Contested Signals (patterns with conflicting interpretations)

Contested signal 1: Burden attribution methods The 5 million deaths figure and associated economic burden estimates embed methodological choices about attributing outcomes to AMR versus comorbidities, health system factors, and confounding variables, with ongoing debates about appropriate attribution frameworks and their implications for priority-setting [11]. Different attribution methods produce substantially different burden estimates, making this a contested rather than settled signal.

Contested signal 2: Governance instrument design Fundamental disagreement over whether AMR requires binding international legal instruments (Framework Convention model) versus strengthened non-binding coordination through existing WHO/FAO/WOAH mechanisms, reflecting deeper tensions about sovereignty, enforcement capacity, and optimal governance architecture [11], [12]. This contestation reveals that the appropriate governance form for AMR remains an open question with significant political and institutional stakes.

Contested signal 3: Evidence-action sequencing Tension between policy analyses arguing that imperfect evidence should not delay action (precautionary principle) versus governance reviews cautioning against policy mimicry without implementation monitoring and adaptive learning, reflecting different epistemologies of policy-making under uncertainty [3], [7], [11]. This contestation concerns the relationship between evidence and action, with implications for how quickly governance should move relative to knowledge accumulation.

Contested signal 4: Agricultural use regulation Disagreement over appropriate regulatory approaches to agricultural antibiotic use, ranging from complete bans on growth promotion to risk-based frameworks allowing continued use under specific conditions, reflecting tensions between animal welfare, food security, and AMR mitigation [3], [29], [30]. This signal is contested because it involves trade-offs among multiple values and stakeholder interests with different governance implications.

Contested signal 5: Pipeline adequacy assessment Conflicting interpretations of whether modest post-2017 antibiotic approvals represent meaningful pipeline progress or continued inadequacy relative to need, depending on how "need" is defined and whether focus is on novel mechanisms versus incremental improvements [20], [21]. This contestation reflects deeper disagreements about what constitutes an adequate R&D response and how to measure pipeline health.

2.4 Signal Set Characteristics and Limitations

This signal set exhibits several systematic characteristics that shape subsequent analysis:

Governance-legibility bias: Signals over-represent phenomena rendered visible by existing governance architectures (NAP adoption, GLASS reporting, WHO priority lists) while under-representing dynamics that escape these measurement regimes (informal antibiotic markets, environmental reservoirs, resistance evolution in under-surveilled regions).

Temporal lag: Most quantitative signals (mortality burden, resistance rates) derive from data collected 2-4 years before publication, meaning 2026 analysis relies substantially on 2022-2024 data, with earlier periods even further back. This lag obscures recent dynamics and creates uncertainty about current state.

Geographic concentration: Signal density highest for high-income countries and WHO-engaged LMICs, with systematic gaps for conflict-affected states, small island nations, and regions with minimal laboratory capacity. The signal set thus describes a partial global picture with known blind spots.

Sectoral asymmetry: Human health signals substantially richer than animal health or environmental signals, reflecting both research concentration and governance capacity differences across One Health domains. Agricultural antibiotic use patterns are asserted more than empirically documented.

Attribution uncertainty: Quantitative burden signals (deaths, costs) embed contested methodological choices about attributing outcomes to AMR versus confounding factors, meaning these signals are more uncertain than their precision suggests.

These characteristics mean the signal set is strongest for analyzing global governance architecture and policy diffusion, moderately strong for clinical resistance trends in surveilled regions, and weakest for

implementation realities, agricultural dynamics, and resistance patterns in under-surveilled areas. The Prismatic Method must process signals aware of these systematic limitations.

Section 3: Grammar Specifications

The Prismatic Method requires processing the signal set through multiple interpretive grammars, each with distinct ontological commitments, causal logics, and closure criteria. This analysis deploys three grammars selected to maximize productive tension and expose how framing shapes interpretation of identical empirical signals.

Grammar 1: Technocratic Governance Optimization

Ontological commitments: This grammar treats AMR as a global public health problem amenable to rational policy intervention through coordinated governance mechanisms. It assumes: (1) resistance is a measurable phenomenon that can be quantified through surveillance; (2) evidence-based interventions (stewardship, infection prevention, R&D incentives) can modify resistance trajectories; (3) international coordination through multilateral institutions (WHO, FAO, WOAH) can align national actions; (4) policy diffusion (NAPs) represents progress toward effective response; (5) implementation gaps are technical problems solvable through capacity-building and resource mobilization.

Causal logic: Problem identification → evidence synthesis → policy formulation → international coordination → national adoption → implementation → outcome measurement → adaptive refinement. Causation flows from knowledge to policy to action, with feedback loops enabling learning and optimization. Governance mechanisms are instruments that, when properly designed and resourced, produce intended effects. Failures reflect insufficient resources, weak capacity, or incomplete implementation rather than fundamental governance limits.

Epistemic framework: Knowledge is produced through scientific research (surveillance, clinical trials, economic modeling) and synthesized into evidence-based guidance by expert bodies (WHO, academic institutions). Policy-makers translate evidence into interventions, and implementation fidelity determines outcomes. Uncertainty is reducible through better data collection and analysis. The grammar privileges quantitative metrics (mortality burden, resistance rates, NAP adoption counts) as objective measures of problem severity and response adequacy.

Temporal orientation: Progressive and optimistic. The 2015 Global Action Plan marks a turning point from neglect to coordinated action. The 2015-2026 period represents first-generation implementation with expected learning curves and capacity-building needs. Future trajectory is improvable through sustained commitment, increased resources, and adaptive governance. Short-term gaps are transitional problems on path to long-term effectiveness.

Spatial logic: Global in scope with national implementation. International institutions set norms and provide technical assistance; nation-states are primary implementation units; subnational and local levels are sites of operational delivery. Geographic disparities (LMIC burden, SSA preparedness deficits) reflect uneven development and resource distribution, addressable through international cooperation and capacity-building. One Health integration requires cross-sectoral coordination within national governance frameworks.

Success criteria: (1) NAP adoption and implementation fidelity; (2) surveillance coverage and data quality; (3) stewardship program deployment and effectiveness; (4) resistance rate stabilization or decline; (5) R&D pipeline adequacy; (6) mortality and morbidity burden reduction; (7) One Health operational integration; (8)

sustainable financing mechanisms. Success is measurable through indicators tracking governance inputs, process measures, and health outcomes.

Failure modes: Insufficient political commitment, inadequate resources, weak technical capacity, poor coordination across sectors or levels of governance, incomplete implementation, competing priorities, external shocks (pandemics, conflicts). Failures are correctable through increased investment, capacity-building, improved coordination mechanisms, and sustained political attention.

Boundary conditions: This grammar operates within assumptions of: functional state capacity, rational policy-making, evidence-policy linkage, implementable interventions, measurable outcomes, and improvable governance. It struggles with: fundamental market failures resistant to incremental incentives, symbolic policy adoption without implementation intent, governance as performance rather than problem-solving, and resistance dynamics that outpace governance response capacity.

Recommended parameters:

- Intervention efficacy threshold: Stewardship and infection prevention interventions must demonstrate ≥20% resistance reduction in controlled settings to warrant scale-up investment.
- Implementation coverage target: ≥70% of healthcare facilities in a jurisdiction must implement core stewardship practices for population-level effectiveness.
- Surveillance adequacy standard: Representative resistance data from ≥80% of population for evidence-based policy-making.

Grammar 2: Political Economy of Pharmaceutical Capital

Ontological commitments: This grammar treats AMR as a crisis of pharmaceutical capitalism, where antibiotic development and use are structured by profit imperatives that systematically misalign with public health needs. It assumes: (1) pharmaceutical R&D is driven by return-on-investment calculations, not health need; (2) antibiotics are uniquely unprofitable due to stewardship (limiting use) and rapid resistance (shortening product lifecycle); (3) market failures are structural features of capitalist medicine, not correctable through incremental incentives; (4) global health governance operates within and reproduces these market logics; (5) AMR burden is distributed according to global inequalities in healthcare access and economic power.

Causal logic: Capital accumulation imperatives → pharmaceutical industry investment decisions → R&D concentration in profitable disease areas → antibiotic pipeline decline → market failure diagnosis → policy responses constrained by capitalist framework → incremental incentives insufficient to overcome structural contradictions → continued industry exit and pipeline inadequacy. Causation flows from economic structure through institutional behavior to health outcomes. Governance interventions that do not challenge underlying market logic reproduce the problem they purport to solve.

Epistemic framework: Knowledge production is shaped by funding sources and institutional interests. Burden-of-disease quantification (5 million deaths, $100-150B costs) translates health crisis into economic terms legible to capital and finance ministries, but this translation obscures distributional questions (who dies, who profits). Surveillance and evidence synthesis are not neutral but reflect priorities of dominant actors (WHO, pharmaceutical industry, high-income country governments). The grammar is skeptical of technocratic claims to objective expertise, viewing them as politically situated.

Temporal orientation: Historical and structural. The current crisis has roots in decades of pharmaceutical industry consolidation, financialization, and shift toward chronic disease markets. The 2015-2026 period represents continued playing-out of structural contradictions, with policy responses (push/pull incentives, subscription models) attempting to preserve market-based R&D while addressing its failures—a fundamentally contradictory project. Future trajectory depends on whether governance challenges or accommodates pharmaceutical capital's logic.

Spatial logic: Global capitalism with uneven development. AMR burden concentrates in Global South due to: weak regulatory capacity enabling uncontrolled antibiotic use, limited access to infection prevention infrastructure, pharmaceutical dumping of substandard drugs, and exclusion from benefits of new drug development (which targets high-income markets). One Health rhetoric obscures that agricultural antibiotic use serves industrial livestock production for global commodity chains. Geographic disparities are not accidental but structured by global political economy.

Success criteria: (1) De-commodification of antibiotic R&D through public pharmaceutical production; (2) delinkage of antibiotic revenues from sales volume; (3) equitable access to existing and new antibiotics regardless of ability to pay; (4) regulation of agricultural antibiotic use that prioritizes public health over livestock industry profits; (5) redistribution of AMR burden away from Global South; (6) democratic governance of pharmaceutical knowledge and production. Success requires structural transformation, not optimization of existing arrangements.

Failure modes: Continued industry exit from antibiotic R&D despite incremental incentives; persistence of access inequities; regulatory capture by pharmaceutical and agricultural industries; symbolic One Health adoption without challenging industrial livestock production; burden remaining concentrated in LMICs; governance reforms that preserve market logic while claiming to address market failure.

Boundary conditions: This grammar operates within assumptions of: capitalism as structuring force, profit-driven pharmaceutical industry, global health governance embedded in market logic, and systematic inequalities in health burden and access. It struggles with: explaining variation in national responses within similar economic structures, accounting for non-market governance innovations, and specifying mechanisms of structural transformation beyond critique of existing arrangements.

Recommended parameters:

- Market failure threshold: When private R&D investment falls below 50% of identified public health need for >5 years, market-based approaches are structurally inadequate.
- Access equity standard: New antibiotics must be available at affordable prices in LMICs within 2 years of approval in high-income markets.
- Delinkage requirement: Antibiotic developer revenues must be decoupled from sales volume through subscription or prize mechanisms to resolve stewardship-profitability contradiction.

Grammar 3: Evolutionary-Ecological Resistance Dynamics

Ontological commitments: This grammar treats AMR as an evolutionary-ecological phenomenon in which microbial populations adapt to selection pressures imposed by antibiotic use across multiple environments (clinical, agricultural, environmental). It assumes: (1) resistance evolution is inevitable given antibiotic use; (2) resistance genes circulate through horizontal gene transfer across bacterial populations and environmental reservoirs; (3) governance interventions modify selection pressures but cannot eliminate evolutionary dynamics; (4) resistance trajectories are path-dependent and may be irreversible; (5) human governance operates on timescales mismatched to microbial evolution.

Causal logic: Antibiotic use (clinical, agricultural, environmental) → selection pressure on bacterial populations → resistance evolution and horizontal gene transfer → resistance gene circulation through environmental reservoirs and host populations → clinical treatment failures → governance responses (stewardship, surveillance) → modified but continued selection pressure → ongoing resistance evolution. Causation is ecological and evolutionary, operating through population-level selection dynamics that governance can modulate but not control. Resistance is not a problem to be solved but a dynamic to be managed within evolutionary constraints.

Epistemic framework: Knowledge comes from microbial genomics, evolutionary biology, and ecological modeling. Surveillance data (resistance rates) are phenotypic manifestations of underlying genotypic

diversity and gene flow networks. Burden quantification (deaths, costs) measures human consequences but obscures microbial dynamics driving those consequences. The grammar privileges understanding resistance mechanisms, transmission pathways, and evolutionary trajectories over governance process measures (NAP adoption, stewardship coverage). Uncertainty is irreducible because evolutionary dynamics are stochastic and path-dependent.

Temporal orientation: Evolutionary and long-term. Resistance genes have circulated for millions of years; antibiotic era (1940s-present) represents brief period of intense selection pressure. The 2015-2026 governance response operates on policy timescales (years to decades) while resistance evolution operates on microbial generation times (minutes to hours) and gene flow timescales (continuous). This temporal mismatch means governance is always reactive, responding to resistance patterns already established. Future trajectory depends on cumulative selection pressure across all antibiotic use domains, with irreversible loss of antibiotic effectiveness as likely outcome.

Spatial logic: Ecological and networked. Resistance genes circulate through: clinical settings (hospitals, clinics), agricultural environments (farms, feedlots), environmental reservoirs (wastewater, soil, water), and host populations (humans, animals, wildlife). One Health is not a policy framework but an ecological reality—resistance genes do not respect sectoral boundaries. Geographic patterns reflect: antibiotic use intensity, sanitation infrastructure, population density, and connectivity through trade and travel. LMICs are not simply "behind" in governance but are sites of intense selection pressure due to unregulated use, weak infection prevention, and environmental contamination.

Success criteria: (1) Slowing resistance evolution through reduced selection pressure (stewardship, infection prevention reducing antibiotic need); (2) containing resistance gene circulation through improved sanitation and environmental management; (3) preserving effectiveness of last-resort antibiotics through strict use restrictions; (4) developing interventions that do not impose selection pressure (vaccines, phage therapy, immunomodulation); (5) accepting that some resistance is irreversible and adapting clinical practice accordingly. Success is relative—slowing rather than stopping resistance evolution.

Failure modes: Continued high-intensity antibiotic use across clinical, agricultural, and environmental domains; inadequate infection prevention allowing transmission; environmental contamination facilitating gene flow; governance focus on surveillance and stewardship without addressing agricultural and environmental reservoirs; development of new antibiotics that simply impose new selection pressures; temporal mismatch between governance response and evolutionary dynamics allowing resistance to become entrenched.

Boundary conditions: This grammar operates within assumptions of: evolutionary biology, ecological connectivity, microbial agency (bacteria as evolving populations, not passive targets), and fundamental limits to human control over microbial dynamics. It struggles with: translating evolutionary insights into actionable governance interventions, accounting for social and political factors shaping antibiotic use, and specifying how governance should operate given irreducible evolutionary uncertainty.

Recommended parameters:

- Selection pressure threshold: Total antibiotic use (human + animal + environmental) must decrease by ≥50% to significantly slow resistance evolution.
- Resistance reversibility window: Once resistance genes reach >10% prevalence in bacterial populations, elimination is evolutionarily implausible; governance must shift to containment.
- Environmental reservoir management: Wastewater treatment must remove ≥99% of antibiotics and resistant bacteria to prevent environmental gene flow amplification.

Grammar Comparison Matrix

Dimension	Technocratic Governance	Political Economy	Evolutionary-Ecological
Primary ontology	Policy problem	Capitalist crisis	Evolutionary dynamic
Causal driver	Knowledge → policy → action	Capital accumulation logic	Selection pressure → evolution
Temporal frame	Progressive (2015 turning point)	Historical-structural	Evolutionary (deep time)
Success metric	Implementation fidelity	Structural transformation	Slowed resistance evolution
Failure explanation	Insufficient resources/capacity	Market logic preservation	Temporal mismatch, irreversibility

These three grammars provide radically different interpretive frameworks for processing identical signals. The Prismatic Method's analytical power derives from maintaining fidelity to each grammar's internal logic while exposing how framing shapes closure—what counts as evidence, what constitutes success, and what interventions appear feasible or necessary.

Section 4: Signal Processing Through Grammars

This section processes the signal set through each grammar in substantive interpretive essays (1000-1500 words each), demonstrating how identical empirical patterns generate different meanings, closures, and action implications depending on interpretive framework.

4.1 Processing Through Technocratic Governance Optimization Grammar

The 2015-2026 period represents a watershed in global AMR governance, marked by transition from fragmented awareness to coordinated international action. The WHO Global Action Plan, adopted at WHA68 in May 2015, established a comprehensive framework encompassing surveillance, stewardship, infection prevention, R&D, and awareness-raising—the five strategic objectives that have structured national and international responses ever since [1], [25]. The subsequent UN General Assembly High-Level Meeting in September 2016 elevated AMR to the rare status of a health issue receiving heads-of-state attention, generating political momentum that cascaded through G7 and G20 processes into national policy-making [1]. This governance architecture represents rational collective action: problem identification through scientific evidence synthesis, policy formulation through multilateral deliberation, and implementation through nationally adapted action plans.

The signal of over 100 countries completing or initiating National Action Plans demonstrates successful norm diffusion through the WHO framework [1], [5], [6]. NAPs serve as the primary governance instrument translating global objectives into national contexts, with most plans incorporating the five strategic objectives and adapting them to local epidemiological patterns, health system capacities, and resource constraints. This rapid policy adoption—most NAPs developed within 2-3 years of the Global Action Plan—reflects effective technical assistance from WHO, FAO, and WOAH, alongside genuine political recognition of AMR as a priority health threat. The tripartite (later quadripartite with UNEP) coordination mechanism institutionalizes One Health principles, ensuring that NAPs address human, animal, and environmental dimensions rather than treating AMR as solely a medical problem [9], [29], [30].

Surveillance expansion through the Global Antimicrobial Resistance and Use Surveillance System (GLASS), launched in 2015, provides the evidence base for informed policy-making [1], [10]. GLASS standardizes resistance monitoring across countries, enabling comparative analysis and identification of emerging threats. While coverage remains incomplete—a recognized challenge addressed through capacity-building initiatives—the system represents substantial progress from the pre-2015 landscape of fragmented, non-comparable national surveillance [10], [11]. Regional networks complement GLASS, providing more granular data for specific geographic contexts. The identification of priority pathogens through the WHO Bacterial Priority Pathogens List, updated in 2024, focuses R&D efforts on organisms posing greatest public health threat, ensuring efficient resource allocation [20].

The quantification of AMR's global burden—approximately 5 million deaths associated with bacterial resistance in 2019—provides crucial evidence for priority-setting and resource mobilization [1], [2]. This figure, derived from comprehensive burden-of-disease modeling synthesizing surveillance data across 204 countries, translates microbial resistance into mortality metrics comparable to other major health threats. The associated economic burden estimates of $100-150 billion annually, with projections of substantially larger long-term costs without intervention, make the case for AMR investment in terms legible to finance ministries and development institutions [2], [9]. These quantitative anchors are essential for competing for resources in crowded global health agendas and justifying the multi-sectoral coordination AMR requires.

Stewardship interventions demonstrate technical efficacy where implemented. The signal of approximately 32% resistance reduction in facilities with documented stewardship programs provides evidence that rational antibiotic use, supported by guidelines, formulary restrictions, and prescriber feedback, can modify resistance trajectories [2]. This effectiveness validates the stewardship pillar of the Global Action Plan and justifies continued investment in stewardship program expansion. Similarly, infection prevention and control measures reduce antibiotic need by preventing infections, addressing AMR's root cause rather than merely managing consequences. The challenge is scaling these proven interventions from pilot sites to system-wide implementation—a capacity-building and resource mobilization problem rather than a technical uncertainty.

The regional disparity signal—3-4× higher resistance rates in LMICs compared to high-income countries—identifies where governance strengthening is most urgent [2]. This pattern reflects multiple reinforcing factors: weaker regulatory oversight enabling uncontrolled antibiotic access, limited laboratory capacity constraining surveillance, under-resourced health systems reducing infection prevention, and competing health priorities limiting AMR-specific investment. The sub-Saharan Africa preparedness deficit, with AMR scores substantially below overall JEE means, pinpoints a region requiring targeted capacity-building [14]. These disparities are not immutable but reflect uneven development addressable through international cooperation, technical assistance, and sustainable financing mechanisms. The Global Action Plan explicitly recognizes that LMIC implementation requires external support, and mechanisms like the Fleming Fund provide resources for surveillance strengthening.

Agricultural antibiotic use, often equaling or exceeding human medical consumption, represents a governance gap requiring attention [3], [29], [30]. The One Health framework recognizes that resistance genes circulate across human, animal, and environmental domains, making siloed interventions insufficient. Many NAPs now include agricultural components, and WOAH standards provide guidance for veterinary antibiotic use. The challenge is implementation, particularly in countries where agricultural sectors have weak regulatory oversight and economic incentives favor continued high-intensity use. This is a coordination problem requiring engagement with agriculture ministries, livestock industries, and veterinary professional bodies—precisely the cross-sectoral collaboration that One Health governance aims to institutionalize.

The pharmaceutical R&D challenge requires innovative incentive mechanisms. The signal of sustained industry exit from antibacterial development despite policy attention reflects fundamental market failures: antibiotics are uniquely unprofitable due to stewardship limiting use and rapid resistance shortening product

lifecycle [9], [21]. Conventional market incentives (patent protection, exclusivity periods) are insufficient. The governance response—push incentives (R&D grants, clinical trial support) and pull incentives (market entry rewards, subscription models)—aims to de-link antibiotic revenues from sales volume, resolving the stewardship-profitability contradiction. The Transatlantic Task Force on Antimicrobial Resistance and other policy forums have developed detailed proposals for subscription models where governments pay fixed fees for antibiotic access regardless of volume used [21]. Implementation of these mechanisms at scale would address the pipeline problem while preserving stewardship principles.

The implementation gap signal—strong vertical NAP alignment with global frameworks but weaker horizontal coordination and operational fidelity—identifies a transitional challenge [4], [5], [7]. First-generation NAPs focused appropriately on policy development and stakeholder engagement. Second-generation implementation requires: operational plans with clear responsibilities and timelines, sustainable financing mechanisms, monitoring and evaluation frameworks enabling adaptive management, and horizontal coordination across ministries and sectors. The characterization of some NAPs as "isomorphic mimicry" is a useful diagnostic, highlighting that policy documents alone are insufficient [7]. The governance response is to strengthen implementation support, develop detailed operational guidance, and establish accountability mechanisms tracking progress beyond policy adoption.

Counter-signals like gonorrhea resistance and continued industry exits are not refutations of the governance approach but indicators of challenge severity and the need for sustained commitment [19], [21]. Gonorrhea resistance demonstrates that some pathogens evolve resistance rapidly even under surveillance, requiring continuous pipeline renewal and treatment guideline updates—precisely what the priority pathogen list and R&D incentive mechanisms aim to provide. Industry exits despite incentive attention suggest that incentive designs require refinement and scaling, not abandonment of the market-based approach. These are problems of insufficient implementation, not fundamental governance failure.

The Covid-19 pandemic, occurring mid-period, both challenged and validated AMR governance architecture [30]. The pandemic strained implementation capacity, diverting resources and attention from AMR programs. However, it also demonstrated the value of surveillance infrastructure, infection prevention capacity, and international coordination mechanisms—all central to AMR response. The pandemic's spotlight on antimicrobial stewardship (given concerns about inappropriate antibiotic use in Covid patients) and One Health approaches (given zoonotic origins) may ultimately strengthen AMR governance by increasing political attention and resource flows.

Looking forward, the governance trajectory is clear: continued NAP implementation with increasing operational fidelity, surveillance expansion addressing geographic gaps, stewardship scaling toward population-level coverage, R&D incentive implementation reviving the pipeline, and One Health integration becoming operational rather than rhetorical. Success requires sustained political commitment, adequate and predictable financing, continued technical assistance for capacity-building, and adaptive learning from implementation experience. The 2015-2026 period established governance architecture; the next decade must deliver implementation at scale. The signals indicate progress is uneven but directionally correct, with identified gaps representing opportunities for targeted intervention rather than fundamental governance inadequacy.

This grammar processes the signal set as evidence of rational governance responding to a complex but manageable global health threat. Policy diffusion, surveillance expansion, stewardship effectiveness, and burden quantification are signs of progress. Implementation gaps, regional disparities, and R&D challenges are transitional problems addressable through capacity-building, resource mobilization, and incentive refinement. The closure this grammar generates is optimistic and actionable: continued commitment to the established governance framework, with adaptive refinements based on implementation learning, will progressively reduce AMR burden and preserve antibiotic effectiveness for future generations.

4.2 Processing Through Political Economy of Pharmaceutical Capital Grammar

The 2015-2026 period reveals AMR as a crisis of pharmaceutical capitalism, where the same market logic that drove antibiotic discovery in the mid-20th century now systematically undermines antibiotic development and equitable access. The WHO Global Action Plan and subsequent governance architecture operate within and reproduce this contradiction, attempting to address market failure through mechanisms that preserve rather than challenge the commodification of essential medicines [1], [25]. The result is a decade of symbolic governance—NAP proliferation, surveillance expansion, stewardship rhetoric—that leaves structural drivers of the crisis fundamentally unchanged.

The pharmaceutical R&D signal is the clearest manifestation of this structural contradiction [9], [21]. Antibiotics are uniquely unprofitable within capitalist medicine because effective AMR governance requires limiting their use (stewardship) and because resistance evolution shortens their commercial lifecycle. A pharmaceutical company investing hundreds of millions in antibacterial R&D faces the prospect that stewardship will constrain sales and resistance will emerge within years, destroying the product's value. This is not a correctable market imperfection but a fundamental incompatibility between public health imperatives and profit maximization. The continued industry exit despite a decade of policy attention to push/pull incentives, subscription models, and de-linkage proposals demonstrates that incremental reforms cannot overcome this structural barrier [21].

The proposed solutions—subscription models where governments pay fixed fees for antibiotic access, market entry rewards, extended exclusivity periods—attempt to make antibiotics profitable while preserving stewardship. But these mechanisms accept the premise that antibiotic R&D must be privately owned and profit-driven, merely seeking to subsidize profitability through public funds. This is socialization of costs (public R&D funding, public procurement) while preserving privatization of knowledge and control (patents, pricing power, production decisions). The Transatlantic Task Force proposals, for instance, recommend subscription payments of $1-2 billion per antibiotic—public money flowing to private firms to solve a crisis created by market logic [21]. This is not market failure correction but market failure accommodation.

The alternative—public pharmaceutical production, where antibiotic R&D and manufacturing are de-commodified and operated as public goods—remains outside the governance conversation. The signal set contains no discussion of public production as a viable model, despite historical precedents (wartime penicillin production, contemporary vaccine manufacturing in countries like Brazil and India) and clear advantages for addressing both pipeline and access problems. This absence is not accidental but reflects the ideological constraints within which global health governance operates: market-based solutions are the only imaginable responses, even to problems created by markets.

The burden quantification signal—5 million deaths, $100-150 billion annual costs—performs specific political work within this framework [1], [2]. Translating microbial resistance into mortality statistics and economic losses makes AMR legible to finance ministries and development institutions, but this translation obscures distributional questions. Who dies? Overwhelmingly, people in LMICs with limited healthcare access. Who profits from antibiotic sales? Pharmaceutical companies headquartered in high-income countries. Who bears the costs of resistance? Health systems in countries least able to afford them. The aggregate burden figures erase these distributional realities, presenting AMR as a universal threat requiring collective action while masking how capitalism structures both the problem and proposed solutions.

The regional disparity signal—3-4× higher resistance in LMICs—is not a technical problem of uneven development but a structured outcome of global pharmaceutical capitalism [2]. LMICs face: unregulated antibiotic markets where substandard and falsified drugs circulate freely, pharmaceutical dumping of products banned or restricted in high-income markets, weak regulatory capacity reflecting broader patterns of state capacity shaped by colonial legacies and structural adjustment, limited access to infection prevention infrastructure due to underinvestment in public health, and exclusion from benefits of new antibiotic development (which targets high-income markets where prices can be sustained). The

technocratic framing treats these as capacity deficits addressable through technical assistance; the political economy framing recognizes them as features of unequal global economic integration.

The sub-Saharan Africa preparedness deficit is particularly stark [14]. SSA bears disproportionate AMR burden while having weakest governance capacity for response—a pattern reproduced across global health domains. The proposed solution—capacity-building through international assistance—maintains dependency relationships where SSA countries remain recipients of technical expertise and resources controlled by institutions in the Global North (WHO, donor agencies, Northern universities). This reproduces colonial patterns of knowledge and resource flows while claiming to address health inequities. Genuine equity would require: technology transfer enabling local antibiotic production, unrestricted access to pharmaceutical knowledge (challenging patent regimes), and redistribution of resources toward SSA health system strengthening—none of which feature prominently in the governance architecture.

The NAP diffusion signal—over 100 countries adopting plans—demonstrates successful norm transmission but reveals the limits of policy isomorphism [1], [5], [6]. NAPs are relatively costless to adopt (requiring primarily document production and stakeholder meetings) but expensive to implement (requiring sustained financing for surveillance, stewardship, infection prevention, and regulatory strengthening). The vertical alignment of NAPs with WHO frameworks alongside weak horizontal coordination and implementation fidelity suggests that policy adoption serves symbolic functions—demonstrating compliance with international norms, accessing technical assistance, and signaling commitment—without necessarily transforming practice [4], [5], [7]. The "isomorphic mimicry" characterization is apt: countries adopt the form of AMR governance without the substance, because substance requires resources and political capital that many governments cannot or will not mobilize [7].

This pattern is not unique to AMR but characteristic of global health governance under neoliberalism: proliferation of plans, frameworks, and partnerships that generate governance activity (meetings, reports, monitoring) without challenging underlying political-economic structures. The Global Action Plan does not question pharmaceutical industry control over R&D, does not challenge industrial livestock production's dependence on antibiotics, does not address how structural adjustment and austerity constrain LMIC health system investment. It operates entirely within the existing order, seeking optimization rather than transformation.

The agricultural antibiotic use signal reveals another dimension of this structural constraint [3], [29], [30]. Antibiotic consumption in livestock often exceeds human medical use, driven by industrial animal agriculture's dependence on antibiotics for growth promotion and disease prevention in crowded, stressful conditions. This is not irrational use by individual farmers but rational response to economic incentives in globalized commodity chains where profit margins are thin and production intensity is high. One Health governance rhetoric calls for reducing agricultural use, but implementation requires confronting powerful agribusiness interests and potentially increasing meat prices—politically difficult in both high-income countries (where consumers resist price increases) and LMICs (where livestock industries are economically important and politically connected).

The stewardship effectiveness signal—32% resistance reduction in implementing facilities—demonstrates that technical interventions work where implemented, but "where implemented" is the crucial qualifier [2]. Stewardship requires: trained personnel (pharmacists, infectious disease specialists), laboratory capacity for diagnostic testing, information systems for monitoring prescribing, and institutional support for enforcing guidelines. These are resource-intensive, meaning stewardship scales primarily in well-resourced settings. The result is a stewardship gap mirroring the broader global health equity gap: effective interventions available where least needed (high-income countries with lower resistance burden) and unavailable where most needed (LMICs with higher burden). This is not a transitional problem but a structural feature of market-based healthcare, where interventions diffuse according to ability to pay rather than health need.

The surveillance expansion signal exhibits similar dynamics [10], [11]. GLASS participation grows, but reporting remains weakest from highest-burden regions due to laboratory capacity constraints. Surveillance

infrastructure is expensive—requiring trained microbiologists, quality-assured laboratories, information systems, and sustained operational funding. International assistance can support surveillance strengthening, but this creates dependency on donor priorities and funding cycles. When donors shift attention (as occurred during Covid-19), surveillance systems strain or collapse. Sustainable surveillance requires domestic resource mobilization, which returns to the fundamental question of health system financing and the fiscal constraints many LMICs face due to debt burdens, tax base limitations, and international financial architecture that prioritizes debt service over social investment.

The counter-signals—gonorrhea resistance, continued industry exits, persistent implementation gaps—are not anomalies but predictable outcomes of attempting to govern a capitalist crisis through capitalist mechanisms [19], [21]. Gonorrhea resistance emerges because evolutionary dynamics operate faster than governance can respond, but also because treatment guidelines must balance effectiveness against cost, and because surveillance gaps mean resistance is often detected only after widespread dissemination. Industry exits continue because subscription models and market entry rewards, even if implemented, do not fundamentally alter the profitability calculus—they merely subsidize unprofitable products rather than transforming the R&D model. Implementation gaps persist because NAPs are unfunded mandates in countries with limited fiscal space and competing priorities.

The Covid-19 pandemic exposed these contradictions sharply [30]. The pandemic demonstrated that pharmaceutical R&D can be rapidly mobilized when political will and public funding align (vaccine development), but also that market-based distribution creates access inequities (vaccine apartheid). It showed that infection prevention and surveillance infrastructure are chronically under-resourced until crisis hits, then rapidly overwhelmed. It revealed that international cooperation rhetoric coexists with vaccine nationalism and intellectual property protection that blocks technology transfer. These lessons apply directly to AMR but remain unintegrated into governance frameworks that continue to rely on market mechanisms and voluntary cooperation.

The closure this grammar generates is pessimistic about incremental reform and insistent on structural transformation. The 2015-2026 governance architecture has produced symbolic progress—plans, surveillance networks, stewardship guidelines—while leaving fundamental drivers unchanged: pharmaceutical R&D controlled by profit-seeking firms, antibiotic access determined by ability to pay, agricultural use driven by industrial livestock economics, and AMR burden concentrated in countries with least capacity to respond. Addressing AMR requires challenging these structures: de-commodifying antibiotic R&D through public production, ensuring equitable access through delinkage and technology transfer, regulating agricultural use even against agribusiness interests, and redistributing resources toward LMIC health systems. Without such transformation, the next decade will likely see continued symbolic governance alongside worsening resistance and persistent inequities—a managed decline rather than effective response.

4.3 Processing Through Evolutionary-Ecological Resistance Dynamics Grammar

The 2015-2026 period represents a brief moment in the deep evolutionary time of bacterial resistance, yet the signals reveal fundamental tensions between governance operating on policy timescales and resistance evolution operating on microbial generation times. Bacteria have evolved resistance mechanisms over millions of years; the antibiotic era (1940s-present) imposed intense selection pressure that accelerated resistance evolution and horizontal gene transfer. The governance response, crystallized in the 2015 Global Action Plan, attempts to slow this evolutionary dynamic through stewardship (reducing selection pressure) and infection prevention (reducing transmission), but the signals suggest governance is structurally reactive, always responding to resistance patterns already established rather than anticipating evolutionary trajectories [1], [25].

The 5 million deaths signal quantifies human consequences but obscures the microbial dynamics generating those consequences [1], [2]. Each death associated with resistant infection represents a clinical

endpoint of complex evolutionary processes: antibiotic exposure selecting for resistant variants, horizontal gene transfer spreading resistance genes across bacterial populations, ecological reservoirs (environmental, animal, human) maintaining resistance gene pools, and transmission networks disseminating resistant strains. Surveillance systems detect resistance phenotypically (organism X resistant to antibiotic Y) but often lack genomic resolution to track resistance gene circulation, transmission pathways, and evolutionary dynamics. This means governance operates with limited visibility into the processes it aims to influence.

The regional disparity signal—3-4× higher resistance in LMICs—reflects differential selection pressure and ecological conditions rather than simply governance gaps [2]. LMICs often have: higher antibiotic consumption (both appropriate and inappropriate), weaker infection prevention enabling transmission, environmental contamination from inadequate sanitation and pharmaceutical manufacturing waste, and higher infectious disease burden creating more opportunities for antibiotic exposure. These factors create intense selection pressure and facilitate resistance gene circulation. The ecological reality is that resistance genes, once established in bacterial populations and environmental reservoirs, are extremely difficult to eliminate. Governance interventions can reduce selection pressure going forward but cannot reverse resistance already entrenched—evolutionary dynamics exhibit hysteresis and path-dependence.

The stewardship effectiveness signal—32% resistance reduction in implementing facilities—demonstrates that reducing antibiotic use can decrease selection pressure and allow susceptible bacterial populations to outcompete resistant ones in specific settings [2]. This is evolutionary ecology in action: when selection pressure (antibiotic exposure) decreases, resistance genes impose fitness costs (maintaining resistance mechanisms requires metabolic resources), allowing susceptible strains to recover. However, this dynamic operates locally and temporarily. Resistance genes persist in low-frequency reservoirs, and reintroduction of selection pressure can rapidly select them back to high frequency. Moreover, facility-level stewardship operates within broader ecological contexts: patients move between facilities, communities, and environments, carrying bacterial populations with them. Resistance gene circulation operates at population and ecosystem scales that exceed individual facility boundaries.

The agricultural antibiotic use signal is ecologically critical because it represents massive, continuous selection pressure across vast bacterial populations in animal hosts and farm environments [3], [29], [30]. Industrial livestock operations concentrate thousands of animals in close proximity, creating ideal conditions for bacterial transmission and resistance gene circulation. Antibiotics used for growth promotion or disease prevention expose enormous bacterial populations to sub-therapeutic concentrations—precisely the conditions that most efficiently select for resistance while minimizing fitness costs. Resistance genes selected in agricultural settings circulate through: direct animal-to-human transmission (occupational exposure, zoonotic pathogens), food chain contamination, environmental pathways (manure application to fields, runoff into water systems), and horizontal gene transfer to human-associated bacteria. One Health is not a policy framework but an ecological reality: resistance genes do not respect sectoral boundaries, and agricultural use creates selection pressure that manifests in human clinical failures.

The environmental dimension, under-represented in the signal set, is ecologically fundamental. Wastewater from hospitals, pharmaceutical manufacturing, and communities contains antibiotics, resistant bacteria, and resistance genes. Wastewater treatment plants, where they exist, are not designed to remove antibiotics or select against resistant bacteria—indeed, they may serve as hotspots for horizontal gene transfer by concentrating diverse bacterial populations under antibiotic selection pressure. Rivers, soils, and aquatic environments become resistance gene reservoirs, with environmental bacteria serving as gene donors to clinically relevant pathogens. The quadripartite expansion to include UNEP recognizes this ecological reality, but governance interventions targeting environmental reservoirs remain minimal [30].

The surveillance expansion signal reveals the challenge of monitoring evolutionary dynamics in real-time [10], [11]. GLASS and regional networks track resistance phenotypes (susceptibility testing results) but provide limited genomic surveillance to identify resistance mechanisms, track gene flow, or detect emerging

resistance before clinical impact. Genomic surveillance, where implemented, reveals that resistance genes circulate globally through travel and trade networks, with novel resistance mechanisms emerging in one region and disseminating worldwide within years. The gonorrhea resistance signal exemplifies this dynamic: ceftriaxone and azithromycin resistance emerged in specific geographic foci and spread through sexual networks, outpacing treatment guideline updates [19]. This is evolutionary dynamics operating faster than governance can respond—by the time resistance is detected, characterized, and incorporated into policy, it has already disseminated.

The temporal mismatch is fundamental. Bacteria reproduce on timescales of minutes to hours; horizontal gene transfer can spread resistance genes across bacterial populations in days to weeks; resistance can emerge and disseminate within months to years. Governance operates on timescales of years (policy development, guideline updates) to decades (infrastructure building, behavior change). This means governance is structurally reactive: surveillance detects resistance after it has emerged, stewardship responds after resistance has become clinically significant, and R&D develops new antibiotics after existing ones have failed. The evolutionary dynamic is always ahead of the governance response.

The R&D pipeline signal must be understood in evolutionary terms [9], [20], [21]. Each new antibiotic imposes new selection pressure, driving evolution of resistance to that antibiotic. The history of antibiotic development is a Red Queen race: new drugs are developed, resistance emerges, newer drugs are developed, resistance emerges again. The WHO Priority Pathogens List identifies organisms for which new antibiotics are most urgently needed—but these are precisely the organisms that have already evolved resistance to multiple existing drugs, demonstrating their evolutionary capacity to respond to selection pressure [20]. Developing new antibiotics without addressing the underlying evolutionary dynamic (continuous high-intensity selection pressure across clinical, agricultural, and environmental domains) simply continues the cycle.

The alternative is interventions that do not impose selection pressure: vaccines preventing infections (no antibiotic exposure), phage therapy using bacterial viruses (different selection pressure than antibiotics), immunomodulation enhancing host defenses, and improved infection prevention reducing transmission. These approaches work with rather than against evolutionary dynamics. However, they receive limited attention in the governance architecture, which remains focused on preserving antibiotic effectiveness through stewardship while developing new antibiotics—a strategy that accepts continued selection pressure as inevitable.

The NAP diffusion signal, from an evolutionary-ecological perspective, is largely irrelevant [1], [5], [6]. Policy documents do not modify selection pressure or gene flow. What matters is actual antibiotic use across all domains (clinical, agricultural, environmental), infection prevention reducing transmission, and sanitation infrastructure limiting environmental reservoirs. The implementation gap signal—strong policy alignment but weak operational fidelity—means that governance activity (NAP development, stakeholder meetings, monitoring frameworks) proceeds while the ecological drivers of resistance (antibiotic use, transmission, environmental contamination) continue largely unchanged [4], [5], [7]. This is governance theater: visible policy activity that does not substantially alter the evolutionary-ecological dynamics driving resistance.

The counter-signals are predictable from an evolutionary perspective. Gonorrhea resistance emerges because Neisseria gonorrhoeae has high mutation rates, efficient horizontal gene transfer mechanisms, and continuous antibiotic exposure through treatment of millions of infections annually [19]. Industry exits from R&D reflect not just market failure but the evolutionary reality that antibiotics have limited effective lifespans—resistance is not a possibility but an inevitability, making antibiotic R&D a poor long-term investment even from a public health perspective if not coupled with dramatic reductions in selection pressure [21]. Implementation gaps persist because reducing antibiotic use requires behavior change across millions of prescribers, patients, and farmers—a social-ecological transformation far more difficult than policy adoption.

The Covid-19 pandemic's impact on AMR governance reveals the fragility of interventions operating against evolutionary dynamics [30]. Pandemic response disrupted stewardship programs, increased empirical antibiotic use (treating suspected bacterial co-infections in Covid patients), strained infection prevention capacity, and diverted resources from AMR surveillance. These disruptions likely increased selection pressure and transmission, accelerating resistance evolution. The evolutionary dynamic is opportunistic: any relaxation of governance pressure allows resistance to advance. This means AMR governance requires sustained, indefinite commitment—there is no endpoint where resistance is "solved" and governance can be relaxed.

The closure this grammar generates is sobering and focused on evolutionary realism. Resistance evolution is inevitable given antibiotic use; the question is not whether resistance will occur but how fast and how extensively. Governance can slow resistance evolution by: dramatically reducing total antibiotic use across all domains (requiring 50%+ reductions, not incremental optimization), improving infection prevention to reduce transmission and antibiotic need, managing environmental reservoirs through improved sanitation and wastewater treatment, and developing non-antibiotic interventions that do not impose selection pressure. Current governance falls far short of these requirements. Stewardship aims for appropriate use rather than minimal use; agricultural use continues at high intensity; environmental reservoirs are largely unaddressed; and R&D focuses on new antibiotics that will face the same evolutionary dynamics as existing ones.

The evolutionary perspective suggests that the 2015-2026 governance response, while well-intentioned, is inadequate to the scale of the evolutionary challenge. Resistance genes are already widely disseminated in bacterial populations and environmental reservoirs; reversing this is evolutionarily implausible. The realistic goal is slowing further resistance evolution and preserving effectiveness of last-resort antibiotics through strict use restrictions—accepting that some resistance is irreversible and adapting clinical practice accordingly. This requires governance to operate on evolutionary timescales (sustained indefinitely) and ecological scales (addressing all selection pressure sources and transmission pathways simultaneously), demands that current governance architecture does not meet. The signals suggest we are managing rather than solving AMR, and even management requires far more aggressive interventions than currently implemented.

Section 5: Closure Comparison Across Six Dimensions

This section systematically compares how each grammar generates closure around six key dimensions, exposing how interpretive framing shapes conclusions drawn from identical signals.

5.1 Problem Definition

Technocratic Governance: AMR is a global public health problem requiring coordinated policy intervention. The problem is defined by quantifiable burden (5 million deaths, $100-150B costs) and characterized by: inadequate surveillance, inappropriate antibiotic use, insufficient infection prevention, declining R&D pipeline, and incomplete One Health integration. The problem is complex but bounded, with clear intervention points (stewardship, surveillance, R&D incentives) and measurable outcomes (resistance rates, mortality burden). Problem severity justifies international coordination and sustained investment, but the problem is fundamentally solvable through rational governance.

Political Economy: AMR is a crisis of pharmaceutical capitalism, where profit-driven R&D and market-based access systematically misalign with public health needs. The problem is defined by structural contradictions: antibiotics are unprofitable precisely when used appropriately (stewardship), resistance evolution destroys product value, and burden concentrates in populations excluded from market-based solutions. The problem extends beyond health to encompass: global inequalities in burden and access,

pharmaceutical industry control over essential medicines, industrial agriculture's dependence on antibiotics, and governance frameworks that accommodate rather than challenge market logic. The problem is structural, requiring transformation of underlying political-economic arrangements.

Evolutionary-Ecological: AMR is an evolutionary-ecological dynamic in which bacterial populations adapt to selection pressure imposed by antibiotic use across multiple environments. The problem is defined by: inevitable resistance evolution given antibiotic use, horizontal gene transfer circulating resistance genes across populations and environments, ecological reservoirs maintaining resistance gene pools, and temporal mismatch between microbial evolution (fast) and governance response (slow). The problem is not solvable in the sense of eliminating resistance, but manageable through reducing selection pressure, limiting transmission, and accepting evolutionary constraints. The problem is fundamentally about human activities (antibiotic use, sanitation, infection prevention) creating ecological conditions that drive resistance evolution.

Comparison: The three grammars define fundamentally different problems. Technocratic governance sees a policy problem amenable to rational intervention; political economy sees a structural crisis requiring systemic transformation; evolutionary-ecological sees an evolutionary dynamic requiring management within biological constraints. These different problem definitions generate different intervention logics and success criteria.

5.2 Causal Attribution

Technocratic Governance: Causation flows from knowledge to policy to action to outcomes. AMR burden results from: inadequate surveillance (knowledge gap), inappropriate antibiotic use (behavior gap), insufficient infection prevention (implementation gap), and declining R&D (market failure). The Global Action Plan addresses these causes through: surveillance strengthening, stewardship programs, infection prevention investment, and R&D incentives. Implementation gaps reflect insufficient resources, weak capacity, or incomplete coordination—correctable through technical assistance and capacity-building. Causation is linear and governable: interventions, when properly designed and implemented, produce intended effects.

Political Economy: Causation flows from economic structure through institutional behavior to health outcomes. AMR burden results from: pharmaceutical capitalism making antibiotics unprofitable, market-based access excluding poor populations, industrial agriculture requiring high-intensity antibiotic use, and global inequalities concentrating burden in LMICs. Policy responses (NAPs, stewardship, R&D incentives) operate within and reproduce these structural causes, attempting to optimize market-based systems rather than transforming them. Implementation gaps reflect not just resource constraints but political-economic barriers: challenging pharmaceutical industry control, regulating agribusiness, and redistributing resources toward LMICs requires confronting powerful interests. Causation is structural and political: governance interventions that do not address underlying power relations and economic logics reproduce the problem.

Evolutionary-Ecological: Causation flows from antibiotic use (selection pressure) through bacterial evolution to resistance emergence and clinical failures. AMR burden results from: high-intensity antibiotic use across clinical, agricultural, and environmental domains creating selection pressure, horizontal gene transfer spreading resistance genes, ecological reservoirs maintaining resistance gene pools, and transmission networks disseminating resistant strains. Governance interventions (stewardship, infection prevention) can modulate selection pressure and transmission but cannot eliminate evolutionary dynamics. Implementation gaps reflect the difficulty of changing antibiotic use behavior across millions of actors and the temporal mismatch between governance (slow) and evolution (fast). Causation is evolutionary and ecological: resistance emerges through population-level selection dynamics that governance can influence but not control.

Comparison: Technocratic governance attributes causation to governable factors (knowledge, behavior, implementation), making intervention straightforward. Political economy attributes causation to structural factors (capitalism, inequality, power), making intervention require systemic transformation. Evolutionary-ecological attributes causation to biological dynamics (evolution, ecology), making intervention require working within biological constraints. These different causal attributions generate different assessments of what interventions are necessary and feasible.

5.3 Evidence Interpretation

Technocratic Governance: Evidence is interpreted as objective input for rational decision-making. The 5 million deaths figure quantifies problem severity; NAP adoption demonstrates policy diffusion success; stewardship effectiveness validates intervention logic; surveillance expansion provides data for evidence-based policy; regional disparities identify where capacity-building is needed. Counter-signals (gonorrhea resistance, industry exits) indicate challenge severity and need for sustained commitment, not governance failure. Contested signals (burden attribution methods, governance instruments) reflect normal scientific debate and policy deliberation. Evidence accumulation progressively reduces uncertainty and enables better-informed interventions.

Political Economy: Evidence is interpreted as politically constructed and serving specific interests. The 5 million deaths figure translates health crisis into economic terms legible to capital and finance ministries, but obscures distributional questions (who dies, who profits). NAP adoption demonstrates symbolic compliance more than substantive transformation. Stewardship effectiveness in well-resourced settings highlights inequitable intervention access. Surveillance expansion maintains dependency relationships where LMICs remain data providers for Northern institutions. Regional disparities reflect structured outcomes of global capitalism, not accidental capacity gaps. Counter-signals reveal contradictions in market-based governance. Evidence is not neutral but reflects the priorities and power of dominant actors.

Evolutionary-Ecological: Evidence is interpreted as partial visibility into complex evolutionary-ecological dynamics. The 5 million deaths figure measures human consequences but obscures microbial processes generating those consequences. Resistance rates are phenotypic manifestations of underlying genotypic diversity and gene flow. Stewardship effectiveness demonstrates local selection pressure reduction but does not address population-level gene circulation. Surveillance provides snapshots of resistance patterns already established, not real-time evolutionary dynamics. Regional disparities reflect differential selection pressure and ecological conditions. Counter-signals (gonorrhea resistance) demonstrate evolutionary dynamics outpacing governance. Evidence is necessarily incomplete because evolutionary processes are stochastic, path-dependent, and operate at scales exceeding governance visibility.

Comparison: Technocratic governance treats evidence as objective and progressively complete; political economy treats evidence as constructed and politically situated; evolutionary-ecological treats evidence as partial visibility into complex dynamics. These different evidence interpretations shape what counts as valid knowledge and what uncertainties are considered reducible versus irreducible.

5.4 Success Criteria and Metrics

Technocratic Governance: Success is measurable through indicators tracking governance inputs, processes, and outcomes: (1) NAP adoption and implementation fidelity; (2) surveillance coverage and data quality; (3) stewardship program deployment and effectiveness; (4) resistance rate stabilization or decline; (5) R&D pipeline adequacy; (6) mortality/morbidity burden reduction; (7) One Health operational integration; (8) sustainable financing. These metrics are quantifiable, comparable across countries, and trackable over time. Success is incremental and cumulative: each indicator improvement represents progress. The 2015-2026 period shows mixed progress (strong NAP adoption, expanding surveillance, persistent implementation gaps), but trajectory is positive with continued commitment.

Political Economy: Success requires structural transformation measurable through: (1) de-commodification of antibiotic R&D through public production; (2) delinkage of revenues from sales volume; (3) equitable access regardless of ability to pay; (4) regulation of agricultural use prioritizing public health over industry profits; (5) redistribution of AMR burden away from Global South; (6) democratic governance of pharmaceutical knowledge. These criteria are qualitative and transformative rather than incremental. The 2015-2026 period shows governance activity (NAPs, surveillance, stewardship) without structural transformation: pharmaceutical R&D remains privately controlled, access remains market-based, agricultural use continues at high intensity, burden remains concentrated in LMICs. By these criteria, the period represents symbolic governance without substantive success.

Evolutionary-Ecological: Success is relative—slowing rather than stopping resistance evolution—measurable through: (1) reduced total antibiotic use across all domains (≥50% reduction); (2) slowed resistance evolution (longer time to resistance emergence for new antibiotics); (3) preserved effectiveness of last-resort antibiotics; (4) reduced transmission through infection prevention; (5) managed environmental reservoirs; (6) development of non-antibiotic interventions. These criteria recognize evolutionary inevitability and focus on slowing dynamics rather than solving problems. The 2015-2026 period shows modest stewardship gains in some settings but continued high-intensity use overall, persistent transmission, largely unaddressed environmental reservoirs, and limited non-antibiotic alternatives. By these criteria, governance is insufficient to substantially slow evolutionary dynamics.

Comparison: Technocratic governance defines success as incremental progress on measurable indicators; political economy defines success as structural transformation; evolutionary-ecological defines success as slowing evolutionary dynamics within biological constraints. These different success criteria generate different assessments of the 2015-2026 period: technocratic sees progress with gaps, political economy sees symbolic governance without transformation, evolutionary-ecological sees insufficient intervention relative to evolutionary challenge.

5.5 Intervention Logic and Feasibility

Technocratic Governance: Interventions are technical and implementable within existing governance frameworks: stewardship programs, surveillance strengthening, infection prevention, R&D incentives, One Health coordination. These interventions are evidence-based (demonstrated effectiveness where implemented), scalable (can be adapted to diverse contexts), and governable (implementable through policy, regulation, and capacity-building). Feasibility constraints are primarily resource availability and technical capacity, addressable through international cooperation and sustained investment. The intervention logic is optimistic: with adequate resources and political commitment, effective response is achievable within current governance architecture.

Political Economy: Interventions require structural transformation challenging existing power relations and economic logics: public pharmaceutical production, delinkage and technology transfer, agricultural use regulation against industry interests, resource redistribution toward LMICs. These interventions are politically difficult (requiring confrontation with pharmaceutical and agribusiness industries), economically disruptive (challenging market-based R&D and access), and ideologically contested (requiring acceptance that market failures are structural, not correctable through incremental reform). Feasibility constraints are primarily political will and power, not technical capacity. The intervention logic is transformative: effective response requires challenging pharmaceutical capitalism, not optimizing it.

Evolutionary-Ecological: Interventions must reduce selection pressure and transmission at population and ecosystem scales: dramatic reductions in total antibiotic use (≥50%), comprehensive infection prevention, environmental reservoir management, non-antibiotic alternatives. These interventions are biologically necessary (required to substantially slow resistance evolution) but socially difficult (requiring behavior change across millions of actors, infrastructure investment, and acceptance of evolutionary constraints). Feasibility constraints are the difficulty of coordinating action at scales matching ecological dynamics and

sustaining interventions indefinitely (no endpoint where governance can be relaxed). The intervention logic is evolutionary-realist: governance must work within biological constraints, accepting that resistance is manageable but not eliminable.

Comparison: Technocratic governance proposes incremental interventions within existing frameworks; political economy proposes structural transformation challenging existing power relations; evolutionary-ecological proposes interventions matching ecological scales and evolutionary timescales. These different intervention logics generate different assessments of what is necessary (incremental optimization vs. structural transformation vs. evolutionary management) and feasible (technically achievable vs. politically difficult vs. biologically constrained).

5.6 Temporal Trajectory and Future Outlook

Technocratic Governance: Trajectory is progressive with expected learning curves. The 2015 Global Action Plan marks a turning point from neglect to coordinated action. The 2015-2026 period represents first-generation implementation with predictable challenges (resource constraints, capacity gaps, coordination difficulties). Future trajectory is improvable through: sustained political commitment, increased and predictable financing, continued technical assistance, adaptive learning from implementation experience, and scaling of proven interventions. The outlook is cautiously optimistic: continued commitment to established governance framework will progressively reduce AMR burden and preserve antibiotic effectiveness. Challenges are transitional, not fundamental.

Political Economy: Trajectory is continuity of structural contradictions with symbolic governance overlay. The 2015-2026 period shows policy proliferation (NAPs, surveillance, stewardship) without transformation of underlying political-economic drivers (pharmaceutical capitalism, global inequalities, industrial agriculture). Future trajectory depends on whether governance challenges or accommodates market logic. Without structural transformation—public production, delinkage, agricultural regulation, resource redistribution—the outlook is continued managed decline: worsening resistance, persistent inequities, and governance activity that performs concern without delivering transformation. Optimism requires political mobilization challenging pharmaceutical and agribusiness power, which current governance architecture does not enable.

Evolutionary-Ecological: Trajectory is continued resistance evolution with governance attempting to slow dynamics it cannot control. The 2015-2026 period shows modest selection pressure reduction in some settings but continued high-intensity use overall, meaning evolutionary dynamics continue largely unabated. Future trajectory depends on whether governance achieves dramatic reductions in total antibiotic use and comprehensive transmission control—neither of which current interventions approach. The outlook is sobering: resistance genes already widely disseminated are evolutionarily difficult to eliminate; new resistance will continue to emerge; governance is structurally reactive, always behind evolutionary dynamics. Realistic goal is slowing resistance evolution and preserving last-resort antibiotics, accepting that some resistance is irreversible. This requires sustained, indefinite governance commitment at scales and intensities exceeding current efforts.

Comparison: Technocratic governance projects progressive improvement; political economy projects continued contradiction without structural transformation; evolutionary-ecological projects continued evolution with governance attempting to slow dynamics. These different temporal trajectories generate different future outlooks: cautious optimism vs. pessimism without transformation vs. evolutionary realism requiring sustained management.

Section 6: Counter-Signals, Leakage, and Stress Test

This section examines signals that resist or complicate the closures generated by each grammar, identifying where interpretive frameworks leak or fail to account for empirical patterns.

6.1 Counter-Signals Challenging Technocratic Governance

Isomorphic mimicry and persistent implementation gaps: The signal of strong NAP vertical alignment with weak horizontal coordination and operational fidelity directly challenges the technocratic assumption that policy adoption represents progress [4], [5], [7]. If NAPs function primarily as symbolic compliance—demonstrating alignment with international norms without transforming practice—then the governance metric of "100+ countries with NAPs" is misleading. The technocratic response treats this as a transitional problem (first-generation plans focused on policy development; second-generation will deliver implementation), but the political economy interpretation (symbolic governance substituting for substantive action) may better explain persistent gaps. This counter-signal suggests that policy diffusion and implementation are decoupled, not sequentially linked as technocratic logic assumes.

Continued industry exit despite incentive attention: The pharmaceutical R&D signal—sustained industry withdrawal despite a decade of policy focus on push/pull incentives, subscription models, and de-linkage proposals—challenges the technocratic assumption that market failures are correctable through incentive design [9], [21]. If incremental incentive reforms (grants, prizes, extended exclusivity, subscription payments) fail to revive the pipeline, this suggests either: (1) incentives are insufficiently generous (technocratic interpretation requiring larger subsidies), or (2) the problem is structural, not correctable through incentives that preserve market-based R&D (political economy interpretation). The continued exits despite policy attention support the structural interpretation, leaking beyond technocratic closure.

Surveillance expansion with persistent blind spots: The signal of growing GLASS participation alongside systematic under-reporting from highest-burden regions challenges the technocratic equation of surveillance expansion with improved evidence base [10], [11], [14]. If surveillance coverage remains inversely correlated with resistance burden—least data from most-affected regions—then expanding surveillance may increase data volume without proportionally improving evidence quality where most needed. This suggests surveillance expansion serves multiple functions beyond evidence generation (demonstrating governance activity, maintaining international cooperation), some of which may be satisfied without achieving comprehensive coverage. The technocratic closure assumes surveillance gaps are transitional capacity problems; the persistent inverse correlation suggests structural barriers.

Stewardship effectiveness versus coverage gap: The signal of strong stewardship effectiveness (32% resistance reduction) in implementing facilities alongside limited coverage, particularly in LMICs and agricultural sectors, challenges the technocratic assumption that demonstrated efficacy translates to population-level impact [2], [11], [24]. If stewardship remains concentrated in well-resourced settings with lower resistance burden, its public health impact is limited regardless of facility-level effectiveness. This is the efficacy-effectiveness gap familiar from implementation science, but the technocratic grammar tends to elide it by treating efficacy evidence as sufficient justification for scale-up without adequately addressing coverage barriers (resources, infrastructure, political economy of agricultural use).

6.2 Counter-Signals Challenging Political Economy

Variation in national responses within similar economic structures: The political economy grammar attributes AMR governance patterns to structural features of pharmaceutical capitalism and global inequalities, but the signal set shows substantial variation in national responses among countries with similar economic structures [5], [6]. Some middle-income countries have achieved relatively strong stewardship implementation and surveillance capacity despite resource constraints, while some high-income countries lag in agricultural antibiotic regulation despite technical capacity. This variation suggests that political will, institutional capacity, and policy entrepreneurship matter beyond structural

economic position. The political economy closure risks over-determining outcomes from economic structure, under-weighting agency and institutional variation.

Non-market governance innovations: The signal set contains examples of governance innovations that operate outside or against market logic: public surveillance networks sharing data as global public goods, open-access research on resistance mechanisms, technology transfer initiatives for diagnostic capacity, and proposals for public pharmaceutical production [10], [24]. While these remain marginal relative to market-based mechanisms, their existence challenges the political economy claim that global health governance operates entirely within and reproduces capitalist logic. These innovations suggest spaces for non-market governance that the political economy grammar, focused on structural constraints, may under-theorize.

Stewardship effectiveness in resource-constrained settings: The signal that stewardship can be effective even in some resource-constrained settings, when adapted to local contexts (e.g., simplified guidelines, task-shifting to available personnel, low-cost diagnostics), challenges the political economy assumption that effective interventions are necessarily resource-intensive and accessible only to well-resourced settings [2], [24]. This suggests that intervention design and adaptation matter, not just resource availability. The political economy closure risks treating resource constraints as absolute barriers, under-weighting the potential for innovation and adaptation.

6.3 Counter-Signals Challenging Evolutionary-Ecological

Stewardship-driven resistance reduction: The signal of 32% resistance reduction in facilities with implemented stewardship challenges the evolutionary-ecological assumption that resistance, once established, is difficult to reverse [2]. This demonstrates that reducing selection pressure can allow susceptible populations to outcompete resistant ones, at least locally and temporarily. While the evolutionary-ecological grammar acknowledges this possibility (fitness costs of resistance), it tends to emphasize irreversibility and path-dependence. The stewardship effectiveness signal suggests that evolutionary dynamics are more reversible than the grammar's pessimistic closure implies, at least at facility scales and short timescales.

Variation in resistance evolution rates: The signal set shows substantial variation in how quickly resistance emerges and disseminates for different pathogen-drug combinations [19], [20]. Some resistances emerge rapidly and spread globally (gonorrhea resistance), while others remain geographically contained or emerge slowly. This variation suggests that evolutionary dynamics are not uniform but depend on: pathogen-specific factors (mutation rates, horizontal gene transfer capacity), drug-specific factors (pharmacokinetics, resistance mechanisms), and ecological factors (transmission networks, use patterns). The evolutionary-ecological grammar tends to treat resistance evolution as inevitable given antibiotic use, but the variation in rates and patterns suggests more contingency than the closure implies.

Effectiveness of non-antibiotic interventions: The signal set includes evidence that vaccines, improved sanitation, and infection prevention can substantially reduce antibiotic use and resistance burden without imposing selection pressure [2], [24]. This demonstrates that interventions working with rather than against evolutionary dynamics can be effective, challenging the evolutionary-ecological grammar's pessimism about governance capacity. While the grammar acknowledges these interventions theoretically, its closure emphasizes the inadequacy of current governance relative to evolutionary challenge, potentially under-weighting the potential of non-antibiotic approaches.

6.4 Leakage: Signals Escaping All Three Grammars

Covid-19 pandemic impacts: The signal of Covid-19 disrupting AMR governance—straining stewardship, increasing empirical antibiotic use, diverting surveillance resources—escapes clean interpretation by any single grammar [30]. Technocratic governance sees it as external shock disrupting implementation (true but

incomplete). Political economy sees it as exposing health system under-resourcing and governance fragility (true but incomplete). Evolutionary-ecological sees it as increasing selection pressure and transmission (true but incomplete). The pandemic's full impact requires integrating: governance disruption (technocratic), structural health system weaknesses (political economy), and evolutionary consequences of increased antibiotic use (evolutionary-ecological). No single grammar captures this complexity.

Conflict-affected states and governance collapse: The signal set notes that conflict-affected countries (Yemen, Syria, Afghanistan) face systematic barriers to AMR response due to disrupted health systems and governance collapse [28]. This escapes all three grammars' primary logics. Technocratic governance assumes functional state capacity; political economy assumes capitalist market structures; evolutionary-ecological assumes some level of governance attempting to manage selection pressure. In conflict settings, these assumptions fail: states are non-functional, markets are disrupted or informal, and governance is absent. AMR dynamics in these settings—likely severe but largely unmeasured—operate outside the observation field all three grammars assume.

Informal antibiotic markets and ungoverned use: The signal set acknowledges but does not deeply engage with informal antibiotic markets, over-the-counter sales without prescription, and other ungoverned use patterns, particularly in LMICs [16]. These dynamics escape governance visibility and intervention. Technocratic governance assumes governable actors (prescribers, dispensers, patients) responsive to guidelines and regulation. Political economy focuses on formal pharmaceutical markets and industry. Evolutionary-ecological focuses on selection pressure but assumes some governance attempting to measure and manage it. Informal markets operate outside these frameworks, creating selection pressure and resistance evolution that governance cannot easily reach.

Microbial agency and evolutionary contingency: All three grammars treat bacteria as objects of governance (technocratic), products of economic structures (political economy), or populations subject to selection (evolutionary-ecological), but none fully theorizes microbial agency—the capacity of bacterial populations to evolve novel resistance mechanisms, horizontal gene transfer networks, and ecological strategies that escape human prediction and control. The evolutionary-ecological grammar comes closest but still frames resistance as response to human-imposed selection pressure. The possibility that microbial evolution operates with autonomy and contingency that exceeds human governance capacity is acknowledged but not fully integrated into any grammar's closure.

6.5 Stress Test: Extreme Scenarios

Scenario 1: Binding international treaty on AMR Suppose a Framework Convention on AMR, modeled on tobacco or climate treaties, is negotiated and enters into force, creating binding obligations for surveillance, stewardship, agricultural regulation, and R&D financing.

- Technocratic response: This would strengthen governance architecture, providing enforcement mechanisms and predictable financing. Success would depend on treaty design (clear obligations, adequate resources, effective monitoring) and implementation fidelity. This scenario fits technocratic logic: stronger governance mechanisms produce better outcomes.
- Political economy response: A binding treaty operating within existing market logic would reproduce structural problems: pharmaceutical R&D would remain privately controlled (unless treaty mandates public production), access would remain market-based (unless treaty requires delinkage), and agricultural use would continue (unless treaty overrides agribusiness interests). The treaty's effectiveness would depend on whether it challenges or accommodates capitalist structures. Symbolic treaty without structural provisions would be isomorphic mimicry at global scale.
- Evolutionary-ecological response: A treaty's impact depends on whether it achieves dramatic reductions in total antibiotic use and comprehensive transmission control. If the treaty mandates incremental stewardship without addressing agricultural use and environmental reservoirs, evolutionary dynamics continue largely unabated. The treaty would need to impose selection

pressure reductions at scales matching ecological dynamics—politically difficult provisions that may be negotiated away.

Scenario 2: Major pharmaceutical company announces viable subscription model Suppose a major pharmaceutical company, with government subscription payments, successfully develops and brings to market a novel antibiotic effective against priority pathogens, demonstrating that subscription models can revive the pipeline.

- Technocratic response: This validates the R&D incentive approach, demonstrating that market failures are correctable through innovative financing. Success would justify scaling subscription models globally, with governments committing predictable payments for antibiotic access. This scenario confirms technocratic optimism about incentive design.
- Political economy response: This represents public subsidy of private pharmaceutical production—socialization of costs while preserving privatization of knowledge and control. The company retains patents, pricing power, and production decisions, with public funds ensuring profitability. This is market failure accommodation, not correction. The scenario confirms political economy critique: market-based R&D requires continuous public subsidy, raising the question of why production should remain private.
- Evolutionary-ecological response: A new antibiotic imposes new selection pressure, driving evolution of resistance to that antibiotic. Without dramatic reductions in total antibiotic use, the new drug simply continues the Red Queen race. The subscription model may solve the market failure but does not address the evolutionary dynamic. The scenario confirms evolutionary-ecological skepticism about R&D-focused solutions without selection pressure reduction.

Scenario 3: Resistance rates stabilize or decline in multiple regions Suppose surveillance data over 5-10 years shows resistance rates stabilizing or declining for multiple pathogen-drug combinations in multiple regions, suggesting governance interventions are working.

- Technocratic response: This would validate the governance approach, demonstrating that stewardship, infection prevention, and surveillance are effective at population scale. Success would justify continued investment and scaling. This scenario confirms technocratic optimism and provides evidence for adaptive refinement of interventions.
- Political economy response: Resistance decline in which regions? If primarily high-income countries with resources for stewardship, this confirms that effective interventions remain inequitably distributed. If LMICs also show decline, this would challenge political economy pessimism, suggesting that governance can deliver substantive outcomes even within capitalist structures. The scenario would require examining whether decline reflects structural transformation or successful adaptation within existing arrangements.
- Evolutionary-ecological response: Resistance decline at what scale and timescale? Facility-level or local declines are consistent with reduced selection pressure allowing susceptible populations to recover, but resistance genes persist in reservoirs and can be reselected. Regional declines sustained over decades would suggest that selection pressure reductions are sufficient to shift evolutionary dynamics—a significant achievement requiring verification that decline is not temporary or geographically limited. The scenario would require genomic surveillance confirming that resistance gene prevalence is declining, not just phenotypic resistance rates.

These stress tests reveal that the three grammars generate different predictions and interpretations even for identical hypothetical scenarios, confirming that closure is grammar-dependent rather than empirically determined.

Section 7: Post-Analysis Reflexive Note

Having processed the signal set through three interpretive grammars, I observe several patterns in how the Prismatic Method operated in this analysis and what it revealed about both the AMR governance field and the method itself.

Interpretive tensions sustained: The method successfully maintained fidelity to each grammar's internal logic, resisting premature synthesis. The technocratic governance grammar generated an optimistic closure emphasizing progress and improvability; the political economy grammar generated a critical closure emphasizing structural contradictions and symbolic governance; the evolutionary-ecological grammar generated a sobering closure emphasizing biological constraints and temporal mismatch. These closures are genuinely incommensurable—they cannot be easily reconciled or averaged. The method's value lies in holding these tensions open, exposing how framing shapes interpretation.

Signal set limitations exposed: Processing signals through multiple grammars revealed systematic gaps in the observation field. All three grammars struggled with: informal antibiotic markets and ungoverned use, environmental reservoirs and gene flow dynamics, conflict-affected states and governance collapse, and microbial agency beyond human control. These gaps are not accidental but reflect the governance-legibility bias of the corpus: phenomena that escape formal governance architectures are under-represented in academic and policy literature. A more complete analysis would require sources beyond the published literature—grey literature, national reports, industry documents, ethnographic studies of antibiotic use practices.

Grammar selection matters: The three grammars were chosen to maximize productive tension: technocratic governance (optimistic, incremental), political economy (critical, structural), and evolutionary-ecological (biological, constraint-focused). Alternative grammar selections would generate different tensions. For example: a global health security grammar (emphasizing AMR as biosecurity threat) would focus on surveillance, outbreak response, and international coordination; a postcolonial grammar would emphasize how AMR governance reproduces colonial knowledge and resource flows; a complexity science grammar would emphasize non-linear dynamics, tipping points, and emergent properties. The Prismatic Method's insights are grammar-dependent, not universal.

Closure comparison as analytical core: Section 5's systematic comparison across six dimensions (problem definition, causal attribution, evidence interpretation, success criteria, intervention logic, temporal trajectory) proved to be the method's analytical heart. This structured comparison made explicit how interpretive framing shapes every aspect of analysis—what counts as the problem, what evidence is relevant, what interventions are feasible, what constitutes success. The comparison reveals that empirical signals underdetermine interpretation; identical data support radically different conclusions depending on framing.

Counter-signals and leakage as epistemic resources: Section 6's examination of signals resisting or escaping each grammar's closure proved epistemically valuable. Counter-signals (isomorphic mimicry challenging technocratic optimism, stewardship effectiveness challenging evolutionary pessimism) prevent grammars from becoming unfalsifiable interpretive schemes. Leakage (Covid-19 impacts, conflict settings, informal markets) identifies phenomena requiring integration across grammars or development of new interpretive frameworks. The method treats anomalies not as noise to be explained away but as signals of interpretive limits.

Methodological validation for Silent Transitions framework: This analysis was framed as methodological validation for R. Jazinski's Silent Transitions framework, which posits that major sociotechnical transformations often proceed through mechanisms invisible to dominant governance logics. The AMR case provides partial validation: the political economy grammar exposes how symbolic governance (NAP proliferation, surveillance expansion, stewardship rhetoric) may proceed without substantive transformation of underlying drivers (pharmaceutical capitalism, agricultural antibiotic use, global inequalities). The gap between policy diffusion and implementation, characterized as "isomorphic mimicry," exemplifies a silent transition—governance activity that performs concern without delivering transformation. However, the evolutionary-ecological grammar suggests an alternative silent transition: resistance evolution proceeding

faster than governance can respond, with microbial dynamics operating outside governance visibility and control. The framework's insight—that dominant logics miss crucial dynamics—is validated, but the specific mechanisms of silent transition are multiple and grammar-dependent.

Reflexive limitations acknowledged: As declared in Section 0, this analysis operates entirely within textual representations produced by specific epistemic communities (academic researchers, WHO officials, policy analysts). I lack embodied experience of AMR's clinical, agricultural, or lived realities. The analysis is strongest on governance architecture and policy dynamics, weakest on ground-truth implementation and microbial evolutionary dynamics. A more complete Prismatic analysis would integrate: ethnographic studies of antibiotic use practices, industry documents on R&D decisions, microbial genomics tracking resistance gene flow, and perspectives from frontline clinicians, farmers, and affected communities. The current analysis is a situated reading of a particular corpus, not a comprehensive account of AMR governance.

Temporal framing effects: The 2015-2026 window, while empirically justified (post-Global Action Plan era), shapes interpretation. Technocratic governance sees this as early implementation with expected learning curves; political economy sees it as continued playing-out of structural contradictions; evolutionary-ecological sees it as a brief moment in deep evolutionary time. A longer temporal frame (1940s antibiotic era to present) would emphasize historical patterns and path-dependencies; a shorter frame (2020-2026, post-Covid) would emphasize recent disruptions and adaptations. The method's insights are temporally situated.

Practical implications uncertain: The Prismatic Method is analytically powerful but practically ambiguous. It exposes how framing shapes closure but does not adjudicate among grammars or prescribe action. A policy-maker reading this analysis might ask: "Which grammar is correct? What should I do?" The method's answer is unsatisfying: all three grammars capture partial truths; action depends on which framing you adopt; choosing a framing is a political and normative decision, not a technical one. This analytical humility is epistemically honest but practically challenging. The method is better suited for critical analysis and methodological validation than for generating actionable policy recommendations.

Future methodological development: This analysis suggests several directions for Prismatic Method refinement: (1) developing protocols for grammar selection to ensure productive tension without arbitrary choice; (2) specifying criteria for when synthesis across grammars is appropriate versus when tensions should remain open; (3) integrating participatory methods to include stakeholder perspectives as grammars rather than treating them as external validation; (4) developing visualization techniques to represent multi-grammar interpretations simultaneously; (5) exploring how the method scales to real-time policy analysis versus retrospective research.

The Prismatic Method (Mark 2) has proven valuable for exposing how interpretive framing shapes closure in the AMR governance field. The method's strength lies in its refusal of premature synthesis and its insistence that identical empirical signals support radically different interpretations depending on ontological commitments, causal logics, and epistemic frameworks. This is both the method's analytical power and its practical limitation: it reveals the constructed nature of closure without providing a meta-framework for choosing among closures. For methodological validation and critical analysis, this is sufficient; for policy prescription, it is deliberately incomplete.

Section 8: Methodological Observations

This final section reflects on the Prismatic Method (Mark 2) as applied to the AMR governance case, offering observations about the method's strengths, limitations, and appropriate use contexts.

8.1 Method Strengths Demonstrated

Exposure of framing effects: The method's core strength is making explicit how interpretive framing shapes every aspect of analysis. By processing identical signals through three grammars with different ontological commitments, the analysis demonstrated that problem definition, causal attribution, evidence interpretation, success criteria, intervention logic, and temporal trajectory are all grammar-dependent. This exposure of framing effects is valuable for: (1) critical analysis of policy discourse (revealing unstated assumptions), (2) interdisciplinary dialogue (making explicit why different disciplines reach different conclusions from same data), and (3) reflexive research practice (acknowledging that all analysis is situated and partial).

Productive tension maintenance: The method successfully resisted premature synthesis, maintaining fidelity to each grammar's internal logic even when grammars generated contradictory closures. This is methodologically difficult—the impulse toward coherence and resolution is strong—but analytically valuable. The tensions among technocratic optimism, political economy critique, and evolutionary realism are not resolvable through better data or more sophisticated analysis; they reflect genuinely different ontological and normative commitments. Holding these tensions open prevents false closure and keeps interpretive options available.

Counter-signal and leakage identification: The method's systematic attention to signals resisting or escaping each grammar's closure (Section 6) proved epistemically valuable. Counter-signals prevent grammars from becoming unfalsifiable by identifying empirical patterns that challenge each grammar's assumptions. Leakage identifies phenomena requiring integration across grammars or development of new frameworks. This treatment of anomalies as epistemic resources rather than noise distinguishes the Prismatic Method from approaches that seek to explain away contradictions.

Structured comparison protocol: The six-dimension closure comparison (Section 5) provided a systematic protocol for exposing how grammars generate different interpretations. This structured approach makes the method replicable and teachable, moving beyond ad hoc interpretive pluralism toward a disciplined comparative methodology. The protocol could be applied to other policy domains (climate governance, pandemic response, digital regulation) with appropriate grammar selection.

8.2 Method Limitations Encountered

Grammar selection arbitrariness: The choice of three specific grammars (technocratic governance, political economy, evolutionary-ecological) was justified by the goal of maximizing productive tension, but this justification is somewhat arbitrary. Alternative grammar selections would generate different insights. The method lacks clear criteria for grammar selection beyond analyst judgment and domain appropriateness. This is not necessarily a flaw—interpretive methods require judgment—but it limits claims to objectivity or comprehensiveness.

Practical ambiguity: The method is analytically powerful but practically ambiguous. It exposes how framing shapes closure but does not adjudicate among grammars or prescribe action. For critical analysis and methodological validation, this is appropriate; for policy-making requiring decisions under uncertainty, it is frustrating. The method reveals that choosing a framing is a political and normative decision, not a technical one, but provides limited guidance for making that choice. This limits the method's direct policy utility while enhancing its critical and pedagogical value.

Resource intensity: The method is time- and expertise-intensive. Processing signals through three grammars with fidelity to each grammar's internal logic requires: deep familiarity with each interpretive tradition, sustained analytical attention, and substantial writing to develop each grammar's closure. The current analysis required extensive engagement with the signal set and enriched paper metadata, plus 8,000+ words of interpretive processing. This resource intensity limits the method's applicability to high-stakes, well-resourced research contexts rather than routine policy analysis.

Corpus dependency: The method's insights are limited by the observation field and signal set. The current analysis was constrained by the corpus's governance-legibility bias, English-language focus, and

under-representation of implementation realities, informal practices, and microbial dynamics. A different corpus would generate different signals and potentially different closures. The method does not overcome corpus limitations but makes them visible through multi-grammar processing. This is valuable for reflexive research but means the method's insights are always corpus-dependent.

Synthesis ambiguity: The method maintains interpretive tensions but provides limited guidance for when synthesis across grammars is appropriate versus when tensions should remain open. In some contexts, integration may be possible (e.g., technocratic interventions addressing political economy concerns through public production models). In others, tensions may be irreducible (e.g., optimism about governance capacity versus evolutionary realism about biological constraints). The method lacks clear criteria for distinguishing these cases, leaving synthesis decisions to analyst judgment.

8.3 Appropriate Use Contexts

Based on this application, the Prismatic Method (Mark 2) is well-suited for:

Methodological validation research: The method is ideal for validating theoretical frameworks (like Jazinski's Silent Transitions) by exposing how dominant governance logics may miss crucial dynamics. The multi-grammar approach reveals what each interpretive tradition sees and misses, supporting meta-theoretical reflection.

Critical policy analysis: The method is valuable for critical analysis of policy discourse, exposing unstated assumptions and revealing how framing shapes problem definition and intervention logic. This is useful for: academic policy analysis, advocacy organizations challenging dominant framings, and reflexive policy-making seeking to understand its own assumptions.

Interdisciplinary dialogue facilitation: The method provides a structured protocol for interdisciplinary teams to make explicit their different interpretive frameworks and explore how those frameworks shape conclusions. This can facilitate productive dialogue across disciplines that typically talk past each other (e.g., public health, economics, evolutionary biology).

High-stakes decision-making under deep uncertainty: When decisions involve deep uncertainty and high stakes (e.g., pandemic response, climate adaptation, emerging technology governance), the method's exposure of framing effects and refusal of premature closure can prevent overconfident decisions based on single interpretive frameworks.

The method is less suited for:

Routine policy analysis: The resource intensity and practical ambiguity make the method impractical for routine policy analysis requiring timely recommendations.

Contexts requiring clear prescriptions: When decision-makers need clear action recommendations, the method's interpretive pluralism and refusal to adjudicate among grammars is frustrating rather than helpful.

Domains with strong empirical consensus: When empirical evidence strongly constrains interpretation (e.g., well-understood technical problems), the method's multi-grammar approach may be unnecessary overhead.

8.4 Relationship to Other Methods

The Prismatic Method shares features with several established approaches while maintaining distinctive characteristics:

Interpretive policy analysis: Like interpretive policy analysis (Yanow, Schwartz-Shea), the Prismatic Method treats policy problems as constructed through framing rather than objective. However, it adds systematic multi-grammar processing and structured closure comparison, moving beyond single-frame interpretive analysis.

Frame analysis: Like frame analysis (Goffman, Entman), the method examines how framing shapes problem definition and solution. However, it operationalizes frames as full interpretive grammars with specified ontological commitments, causal logics, and epistemic frameworks, providing more systematic comparison.

Triangulation: Like triangulation methods, the Prismatic Method uses multiple perspectives to enhance validity. However, it does not assume perspectives converge on a single truth but maintains that different grammars generate genuinely different closures that may be incommensurable.

Critical realism: Like critical realism (Bhaskar), the method distinguishes between empirical observations (signals) and interpretive frameworks (grammars), acknowledging that reality is stratified and theory-laden. However, it is more agnostic about ontological depth, treating grammars as interpretive tools rather than claims about underlying causal mechanisms.

Poststructural policy analysis: Like poststructural approaches (Bacchi, Fischer), the method is skeptical of objective problem definition and attentive to power and discourse. However, it maintains more structured analytical protocols and does not fully embrace poststructural epistemology.

The Prismatic Method occupies a methodological space between interpretive pluralism and structured comparison, offering a disciplined approach to multi-frame analysis while acknowledging the constructed and political nature of closure.

8.5 Future Development Directions

This application suggests several directions for method refinement:

Grammar selection protocols: Develop clearer criteria for grammar selection, balancing: domain appropriateness (grammars should be relevant to the phenomenon), productive tension (grammars should generate different closures), and stakeholder representation (grammars should reflect perspectives of affected communities). This could involve participatory grammar selection with stakeholders.

Synthesis criteria: Specify conditions under which synthesis across grammars is appropriate versus when tensions should remain open. This might involve: empirical criteria (when grammars converge on similar interpretations), normative criteria (when synthesis serves justice or equity goals), or pragmatic criteria (when decision-making requires integration).

Participatory adaptations: Explore participatory versions where stakeholders contribute grammars based on their lived experience rather than analyst selecting grammars from academic traditions. This would democratize the method and ensure grammars reflect diverse perspectives.

Visualization techniques: Develop visual representations of multi-grammar interpretations (e.g., parallel timelines showing different temporal trajectories, network diagrams showing different causal attributions, matrices comparing success criteria). This would make multi-grammar insights more accessible.

Real-time application: Adapt the method for real-time policy analysis rather than retrospective research. This would require: rapid grammar deployment, abbreviated processing, and integration with decision-making timelines. The trade-off between analytical depth and timeliness would need careful management.

Computational support: Explore computational tools to support signal processing through grammars, particularly for large corpora. Natural language processing could identify signals consistent or inconsistent with each grammar's assumptions, though interpretive depth would remain human-dependent.

8.6 Concluding Methodological Reflection

The Prismatic Method (Mark 2) is a valuable tool for exposing how interpretive framing shapes closure in complex policy domains. Its strength lies in systematic multi-grammar processing that makes explicit what is usually implicit: that problem definition, evidence interpretation, and intervention logic are grammar-dependent, not empirically determined. This exposure of framing effects serves critical, pedagogical, and reflexive purposes, enhancing awareness of how analysis is always situated and partial.

The method's limitation is its practical ambiguity—it reveals interpretive pluralism without providing a meta-framework for choosing among interpretations. This is epistemically honest but pragmatically challenging. The method is best understood as a critical and reflexive tool rather than a prescriptive one, suited for contexts where exposing assumptions and maintaining interpretive options is more valuable than generating clear recommendations.

For the AMR governance case, the method successfully demonstrated that identical signals (5 million deaths, NAP adoption, stewardship effectiveness, R&D decline, regional disparities) support radically different closures depending on whether one adopts technocratic governance, political economy, or evolutionary-ecological framing. This is not a failure of evidence or analysis but a fundamental feature of complex policy domains: empirical signals underdetermine interpretation, and closure requires normative and political choices about which framing to adopt.

The Prismatic Method makes these choices visible and contestable rather than hidden and naturalized. In doing so, it contributes to more reflexive, pluralistic, and democratically accountable policy analysis—not by providing answers but by exposing the questions that must be asked before answers can be given.

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TL;DR

Global estimates place ~5 million deaths linked to bacterial AMR in 2019 with marked excess burden and resistance in low- and middle-income countries. Surveillance, stewardship, One Health adoption and R&D present mixed signals: policy alignment is common but implementation, financing, and surveillance coverage remain uneven.

Global burden patterns

This section summarizes empirical mortality/morbidity signals and major regional differences relevant for 2015–2026 analyses. It highlights what the supplied literature directly reports and what the corpus does not provide in detail.

Metric	Estimate or pattern	Source
Global deaths associated with bacterial AMR (2019)	~5 million deaths	1 2
Relative resistance burden by income group	Resistance rates and burden substantially higher in low- and middle-income countries (reported as ~3–4× higher resistance rates versus high-income countries)	2
Age and population patterns	Older adults and children noted as higher-risk groups in descriptive syntheses, but comprehensive global age-stratified morbidity incidence not consistently reported	insufficient evidence

- Key quantitative signal Global attributable/associated deaths in 2019 are reported at roughly five million in major syntheses
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  .
- Regional disparity signal Low- and middle-income countries bear a disproportionate share of resistance and associated deaths, with reported resistance rates several-fold higher than in high-income settings

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- Data gaps There is no consistent, multi-source, time-series morbidity incidence breakdown (e.g., DALYs or age-specific clinical morbidity) across 2015–2026 in the supplied corpus; therefore fine-grained morbidity trend estimation is not supported by these documents.

Use trends human versus agriculture and stewardship

This section compiles evidence on antibiotic consumption trends across human and animal/agricultural sectors, national stewardship uptake, and One Health adoption signals. It emphasizes measured implementation versus policy presence.

Antibiotic consumption and agricultural use

- Pattern signal Several policy and review sources report that antibiotic use in agriculture often equals or exceeds human use in many countries, and agricultural use is a major AMR driver cited across documents
[3]
.
- Regional/income patterns Surveillance and preparedness analyses indicate higher unregulated or less-regulated veterinary/agricultural use and associated governance weaknesses in lower-income regions, especially sub-Saharan Africa and parts of Asia
[4]

[3]
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- Measurement gap Quantitative, comparable regional time-series (2015–2026) disaggregating human versus animal tonnes or defined daily doses by region/country are not consistently provided in the supplied corpus; therefore precise trend tables by region cannot be constructed from these sources alone.

Stewardship implementation and effectiveness

- Adoption signal Most countries have produced national AMR action plans since WHO's 2015 Global Action Plan, with many countries reporting NAP development or near completion in the years following 2015
[5]

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- Effectiveness signal Facility-level antimicrobial stewardship interventions are reported to reduce resistance in participating sites; one large synthesis reports stewardship programs reduced resistance rates by about 32% where implemented
[2]
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- Implementation gap Cross-country governance assessment finds strong policy alignment (NAP text) but substantially lower implementation fidelity and horizontal coordination, a pattern described as "isomorphic mimicry" where plans exist but operationalization lags
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One Health adoption

- Policy adoption One Health is embedded in global guidance and most NAPs, and vertical alignment with the WHO Global Action Plan is common (particularly among lower-income countries), but horizontal and operational integration across sectors is incomplete
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- Operational gap Real multisectoral One Health implementation (joint surveillance, integrated stewardship across human/animal/environmental sectors) shows wide variation and is generally weaker where AMR preparedness scores and laboratory capacities are low
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Pipeline dynamics economic incentives and market signals

This section extracts empirical signals about antibiotic R&D, approvals, investment trends, market failures, and modeled economic impacts from the supplied literature.

Antibiotic R&D and approvals

- Approval signal Recent priority-pathogen reviews indicate a modest flow of new agents: at least a detectable but limited number of new antibiotics targeting priority pathogens were approved in the post-2017 period as noted in policy/prioritisation updates
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- R&D constraint signal Multiple reviews and task-force analyses document sustained decline in conventional pharmaceutical investment into new antibacterials and highlight the need for combined push/pull or de-linkage incentives to revive the pipeline
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Pharmaceutical investment and market failures

- Market failure signal Economic analyses and policy reviews identify fundamental market failures for antibiotics (low returns, stewardship-driven constrained sales), prompting proposals for subscription and de-linkage models; these reviews argue for coordinated incentives to re-attract industry
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- Observed disinvestment The literature documents withdrawal of investor interest and consolidation in antibiotic R&D over prior decades, though specific firm-level quantitative capital flows for 2015–2026 are not enumerated in the supplied corpus
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Economic cost modeling

- WHO/academic estimates Recent synthesis work in the corpus estimates large annual economic burdens (reported ranges in the supplied review approximate US$100–150 billion annual costs with higher projected aggregates by 2030 in model scenarios) and cites macroeconomic scenarios warning of much larger long-term losses without action
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- Policy implication These economic signals are repeatedly used to justify policy incentives and international coordination mechanisms to correct market failures
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- Data limits on pipeline detail The supplied papers do not provide a comprehensive, publicly accessible register within this corpus listing all active clinical-stage antibacterial candidates, trial counts, or systematic counts of market withdrawals across 2015–2026; therefore a complete empirical pipeline inventory cannot be produced from these documents alone.

Surveillance quality policy gaps counter-signals and contested evidence

This section synthesizes surveillance coverage, implementation gaps between income groups, documented intervention failures or surprising resistance trends, and areas of contested interpretation present in the corpus.

Surveillance coverage and quality disparities

- Coverage signal WHO's GLASS and regional networks exist but reporting is uneven; many low- and middle-income countries have limited laboratory capacity and under-reporting, producing major geographic gaps in surveillance data
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- Preparedness signal Sub-Saharan Africa shows notably lower AMR preparedness scores compared with other JEE indicators; mean AMR preparedness in SSA was substantially below overall JEE means, reflecting laboratory, IPC and NAP implementation shortfalls
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Policy implementation gaps high versus low income

- Alignment vs implementation Analyses show strong textual alignment of NAPs with the 2015 Global Action Plan across many countries (vertical alignment) but weaker horizontal/regional coordination and lower implementation fidelity, particularly in operationalizing recommended actions in lower-resource settings

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- Resource constraint signal Reviews emphasize that resource scarcity, weak health systems, and competing priorities constrain operational uptake of stewardship, surveillance, and animal-sector reforms in LMICs
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Counter-signals and intervention failures

- Unexpected resistance patterns Sentinels of clinical failure include emerging ceftriaxone- and azithromycin-resistant Neisseria gonorrhoeae documented in global surveillance reporting, underscoring rapid clinically consequential resistance emergence despite stewardship efforts
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- Policy-practice disconnect Cases of “isomorphic mimicry” where NAPs exist but actions lag are repeatedly noted as counter-signals to claims of widespread effective implementation
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- Market failure consequences The persistent industry exit from antibiotic development despite policy attention is itself a counter-signal to optimism about rapid pipeline renewal without structural incentive changes
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Contested evidence and measurement disputes

- Governance and binding instruments Debates in the literature focus on whether binding international legal instruments are needed versus strengthened non-binding coordination; tradeoffs and governance design remain contested
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- Causation and attribution Several syntheses note gaps and methodological variability in attributing clinical outcomes directly to AMR versus comorbidity or health-system factors, producing debate over precise burden attribution methods and priorities for surveillance metrics
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- Evidence-action tension Some policy analyses argue that lack of perfect evidence should not delay action, while governance reviews caution against policy mimicry without implementation monitoring; the literature thus contains normative disagreement on sequencing and types of interventions
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- Areas needing targeted empirical work High‑quality, comparable time‑series on human versus animal antibiotic tonnage by country; detailed clinical morbidity (DALYs) attributable to AMR by region; an open, up‑to‑date inventory of clinical‑stage antibacterial assets and firm financing flows for 2015–2026 are not fully available within the supplied corpus and would materially strengthen empirical AMR analyses.

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